Data-Parallel Deep Learning

Efficient DL, Episode 3, 2026

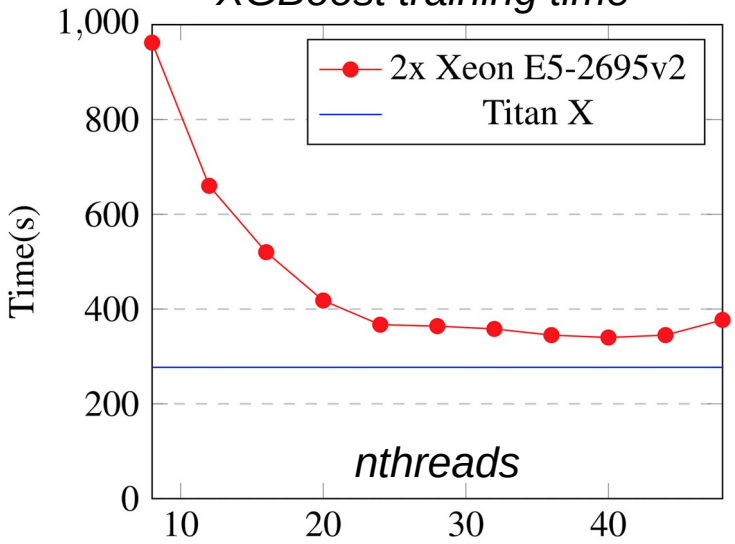
Yandex
Research

LAMBDA

British Hedgehog
Preservation Society

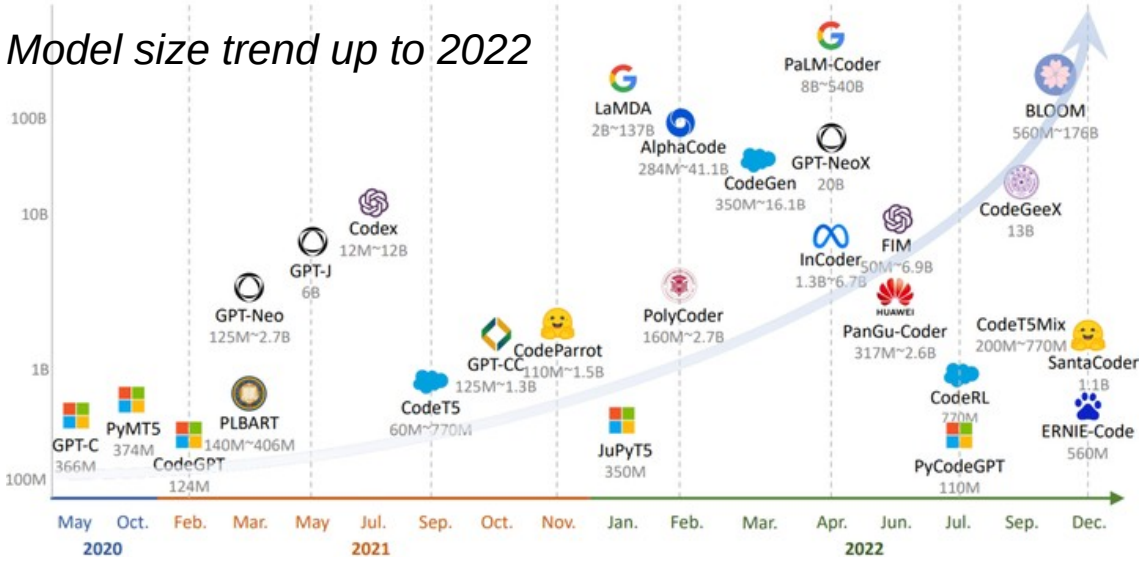
Зачем это всё?

XGBoost training time



parameters vs time

Model size trend up to 2022



BERT-Large Training Times on GPUs

Time	System	Number of Nodes	Number of V100 GPUs
47 min	DGX SuperPOD	92 x DGX-2H	1,472
67 min	DGX SuperPOD	64 x DGX-2H	1,024
236 min	DGX SuperPOD	16 x DGX-2H	256

*(single GPU – **over 2 weeks**)*

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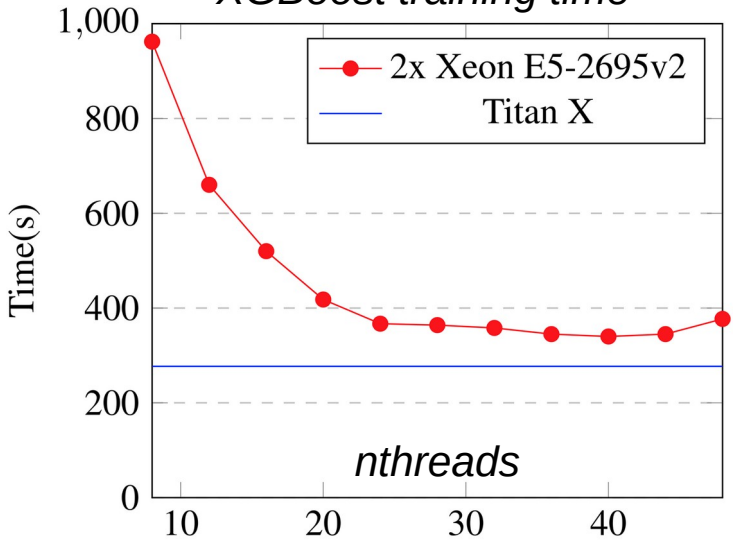
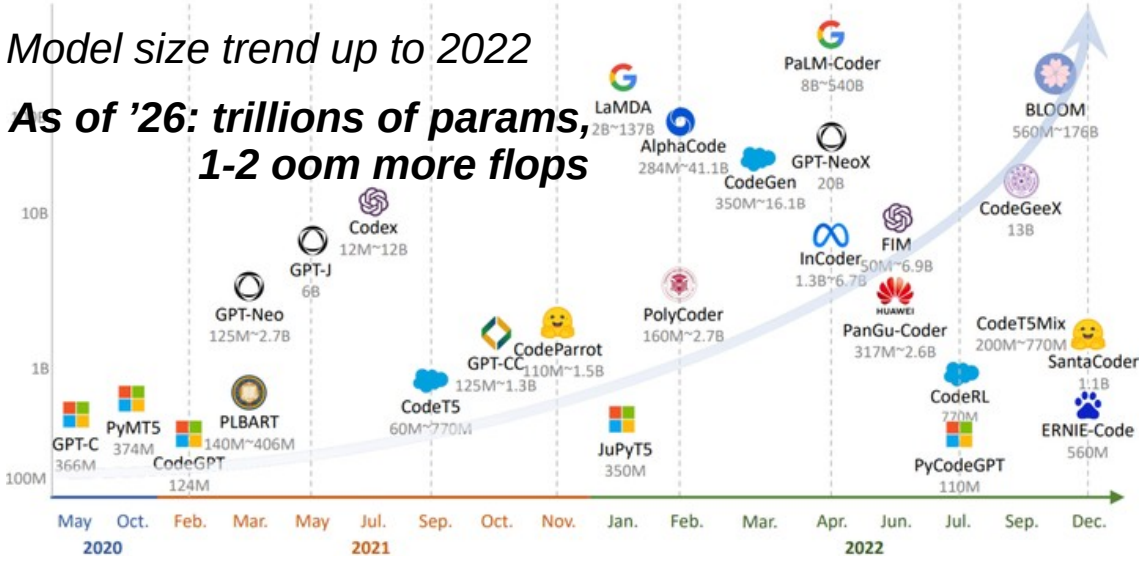


parameters vs time

Model size trend up to 2022

As of '26: trillions of params, 1-2 oom more flops



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Зачем мы тут?

Заставить много железяк вместе учить одну модель

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Заставить много железяк вместе учить одну модель

понять общие подходы

закодить своими руками

на python / pytorch

TL;DR our plan

next few lectures

- Data-parallel deep learning

  Train BERT-base on wikipedia in 20 minutes or less

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  Fine-tune and deploy models with 100B+ parameters

TL;DR our plan

next few lectures

- Data-parallel deep learning

  Train BERT-base on wikipedia in 20 minutes or less

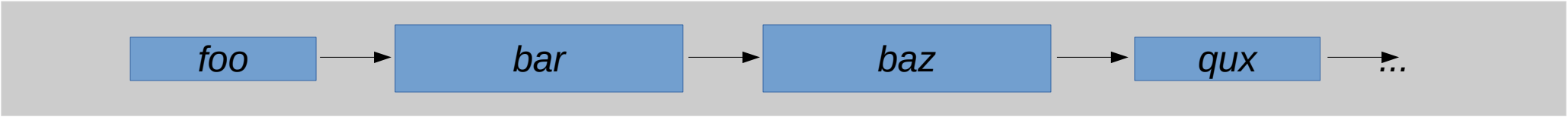
- Model-parallel deep learning

  Fine-tune and deploy models with 100B+ parameters like OPT, Llama, Qwen, DeepSeek R1, ...

- Advanced techniques

  Sharding (FSDP), mixed / hybrid parallelism, practice

Rules: Process



Process:

- Runs some code
- Has some memory
- No one else can access your memory

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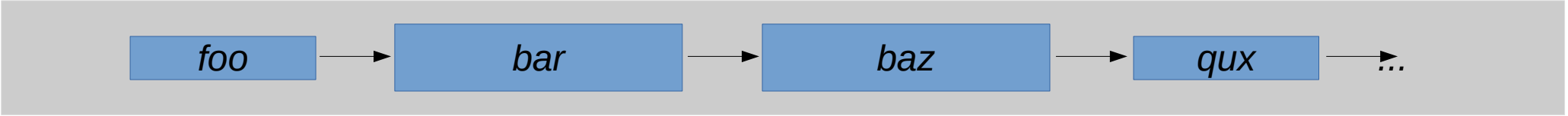

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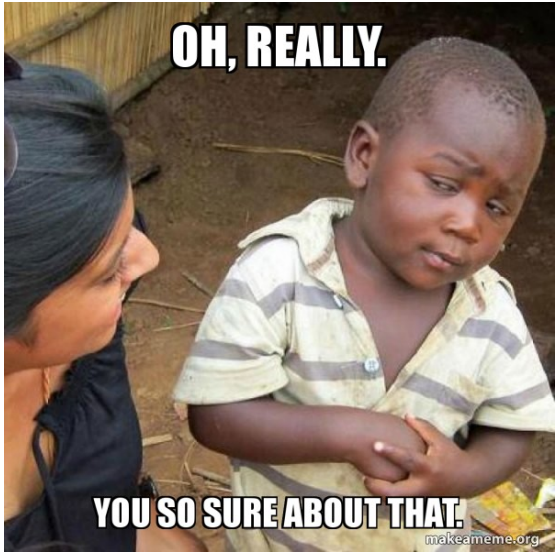

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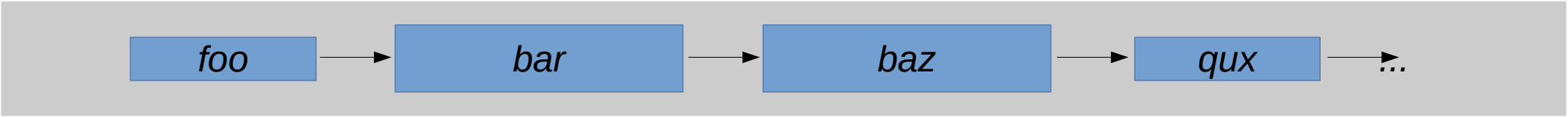

Process:

- Runs some code
- Has some memory
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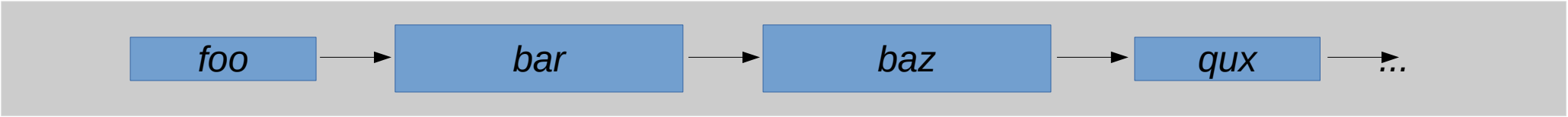
* – not if you use shared memory

Rules: Process



Process:

- Runs some code
- Has some memory
- No one else can access your memory*†

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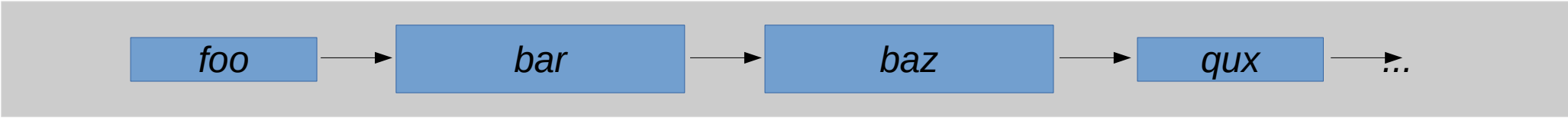
† – superuser can still do that (os-dependent)

Rules: Process

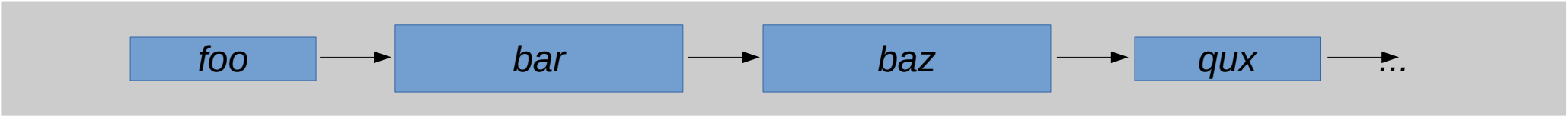


Process:

- Runs some code
- Has some memory
- No one else can access your memory*†‡

* – not if you use shared memory
† – superuser can still do that (os-dependent)
‡ – attacker can do that through spectre/meltdown/etc

Rules: Process

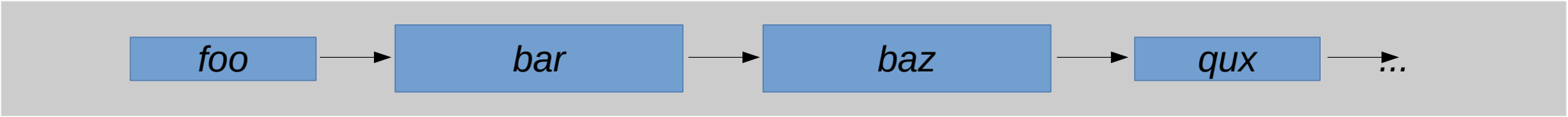


Process:

- Runs some code
- Has some memory
- No one else **should** access your memory*†‡

*†‡ – not relevant for this course

Rules: Process



Process:

- Runs some code
- Has some memory
- No one else **should** access your memory*†‡

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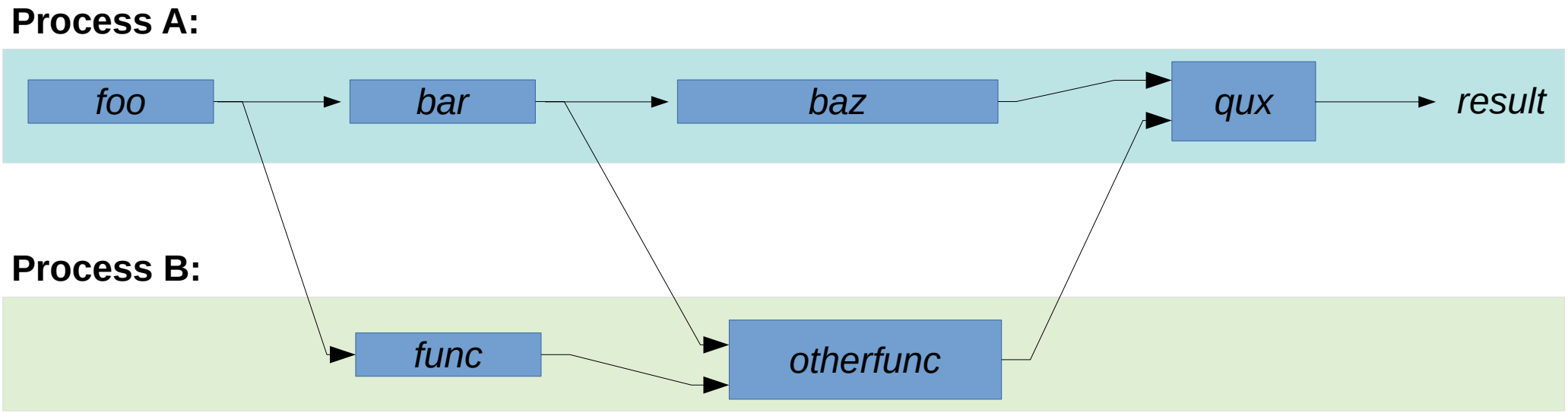
Q: How do we make processes work together?

Rules: Channel / Pipe



Channel (pipe):

- Communication in O(message size)
- Asynchronous read/write

Data-parallel training (naive)

cs.cmu.edu/~muli/file/parameter_server_osdi14.pdf

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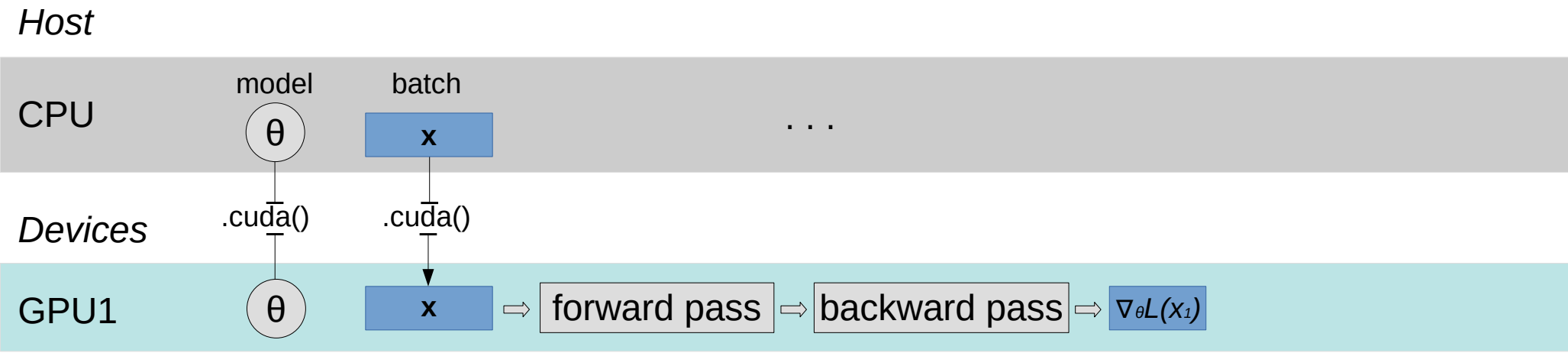

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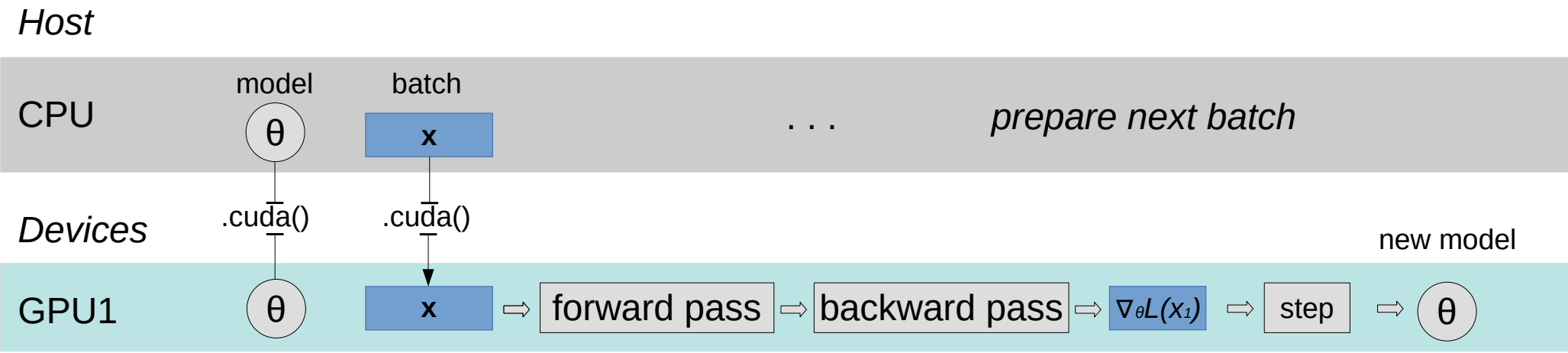

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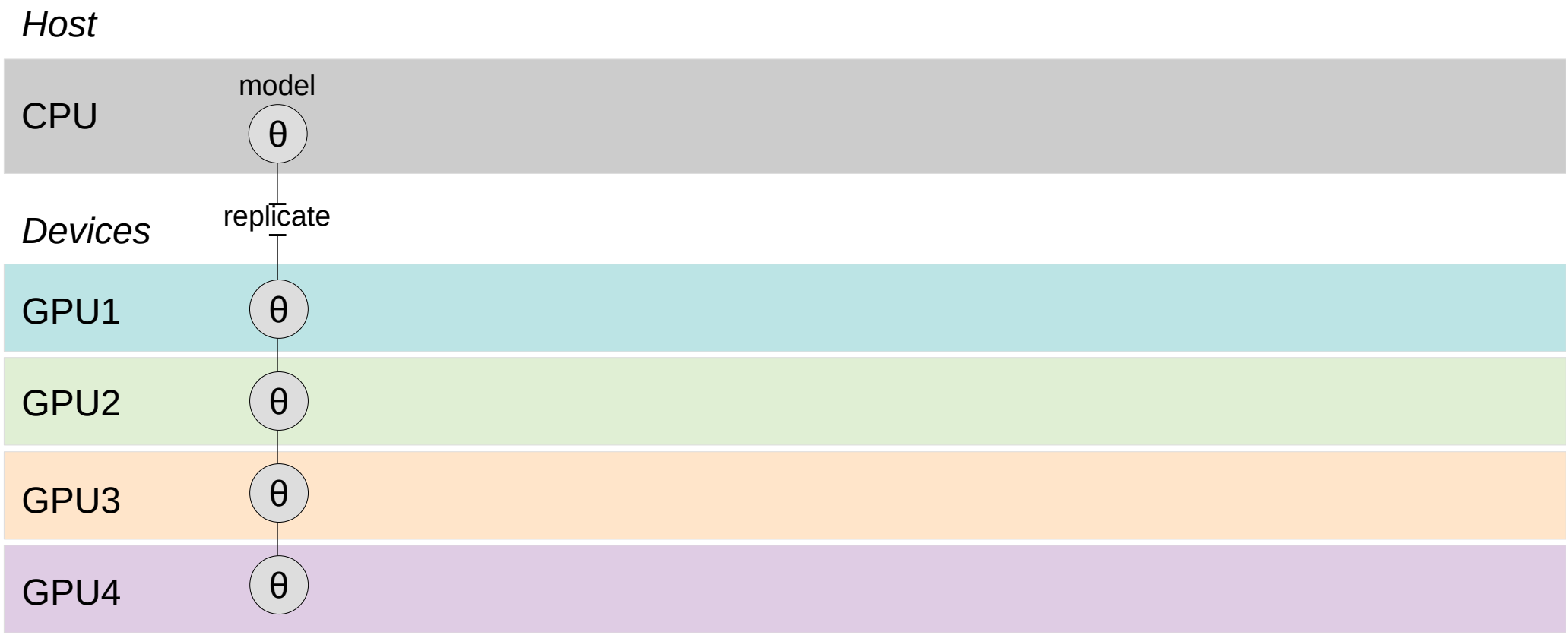

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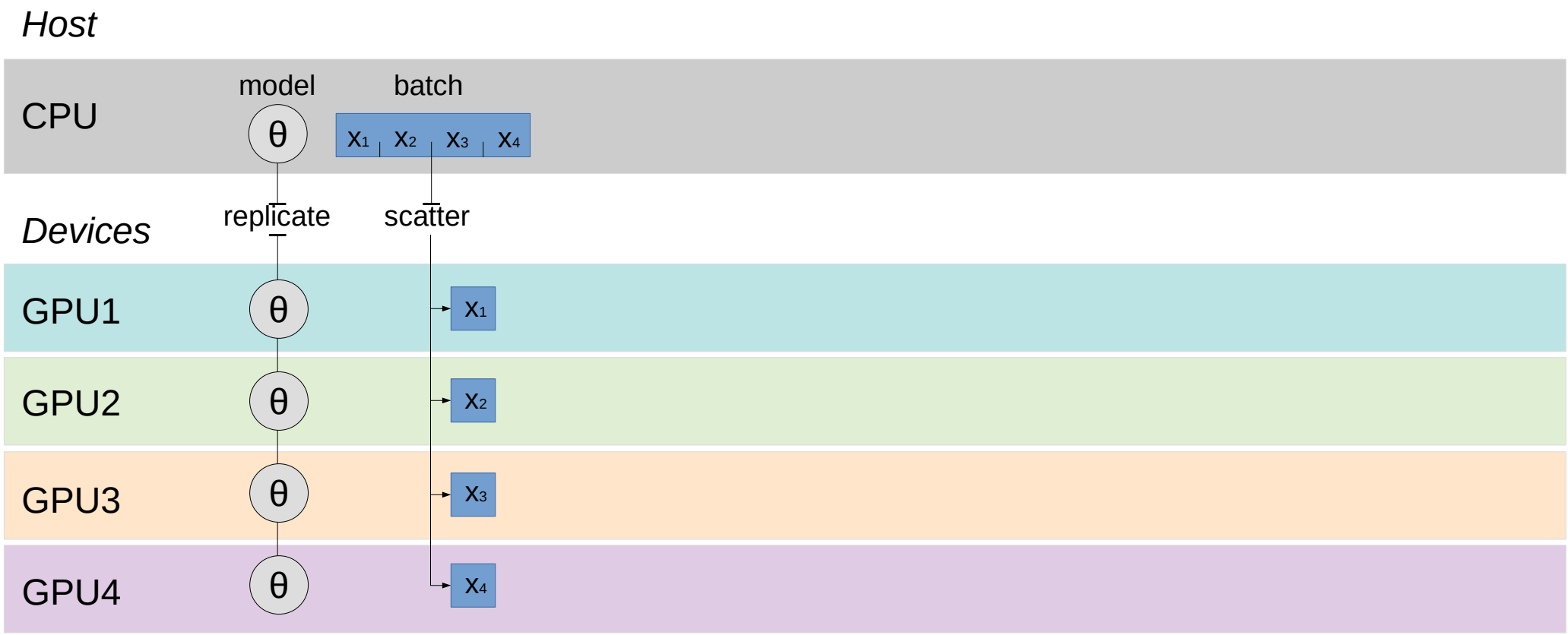

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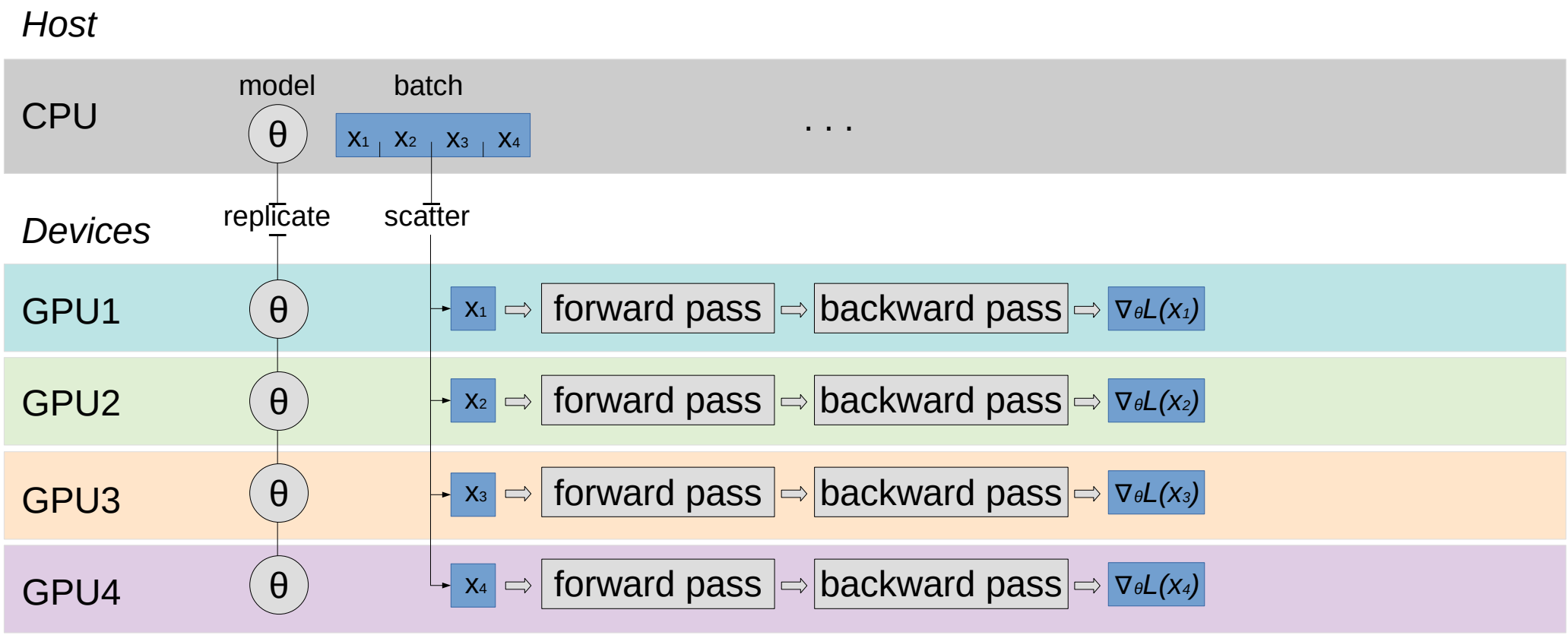

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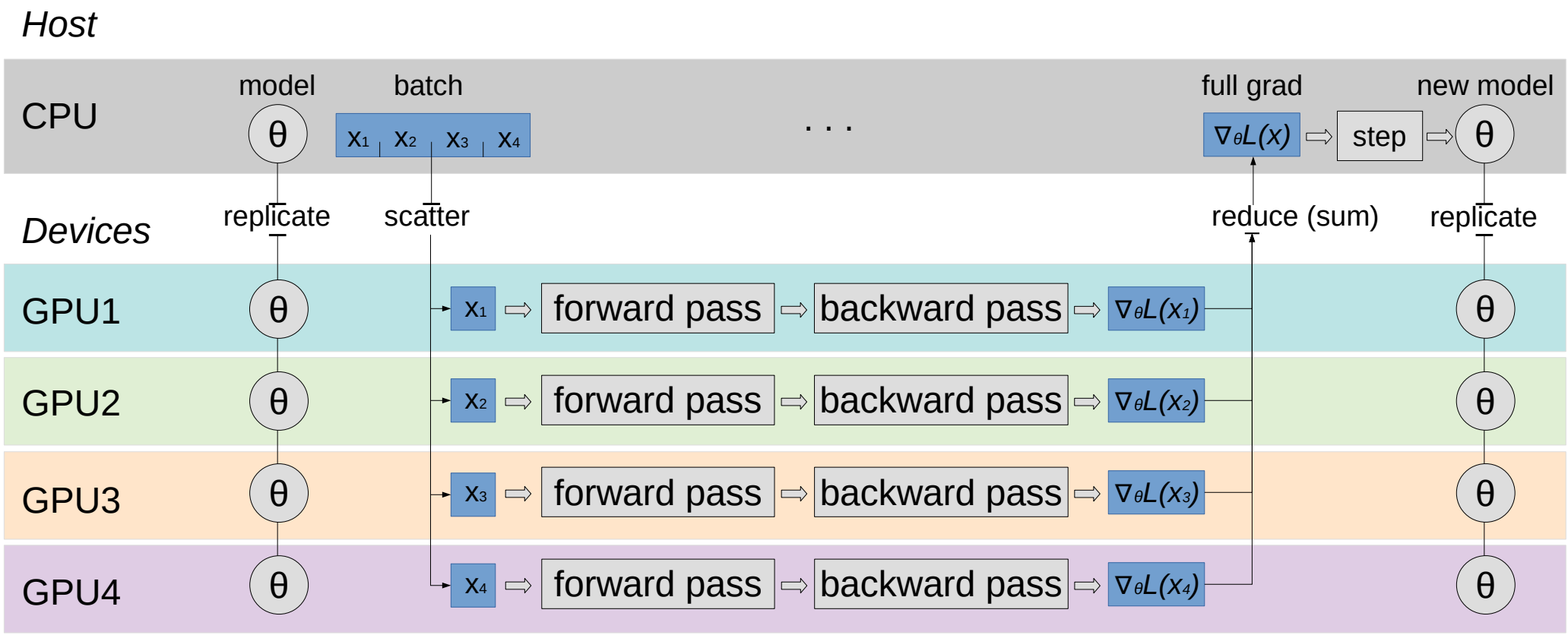
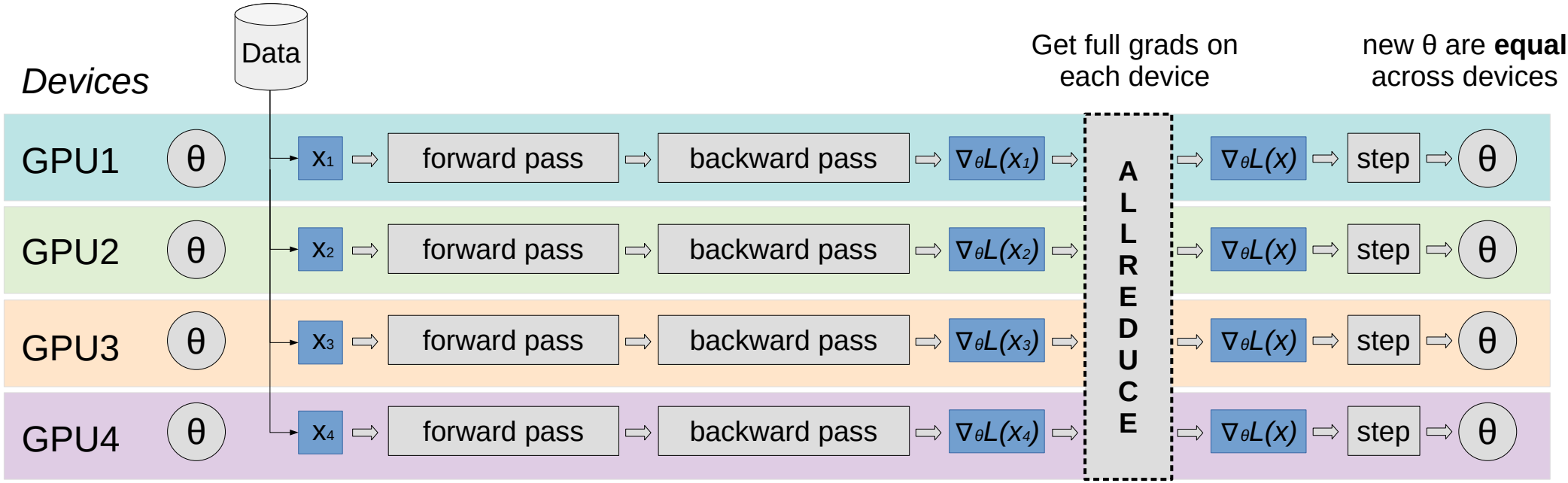


All-Reduce data parallel

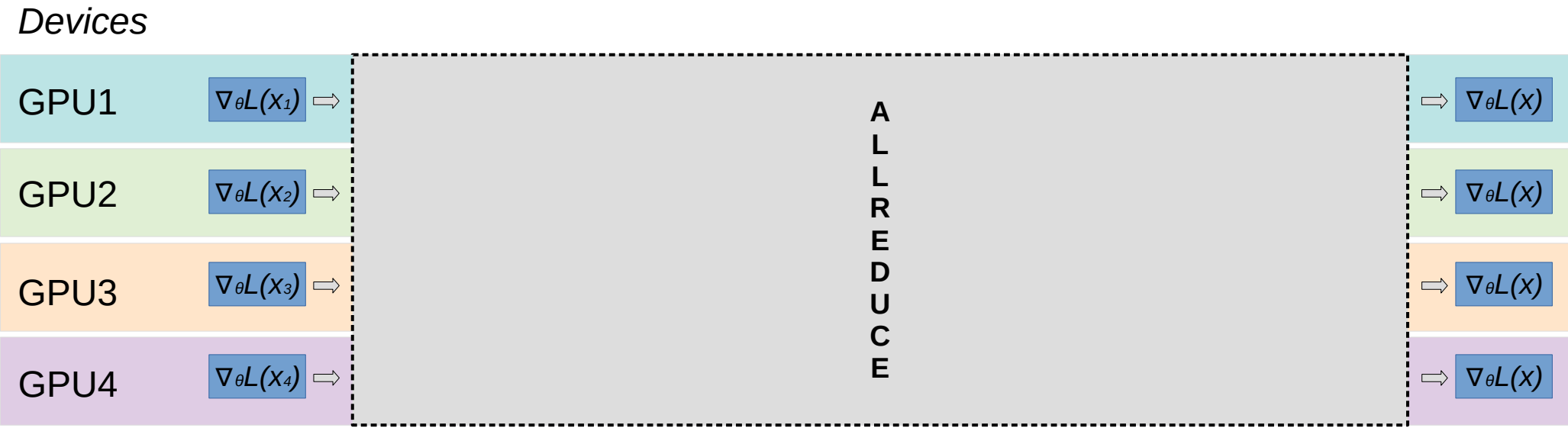
arxiv.org/abs/1706.02677

Idea: get rid of the host, each gpu runs its own computation
Q: why will weights be equal after such step?



Faster allreduce

Input: each device has its its own vector
Output: each device gets a sum of all vectors



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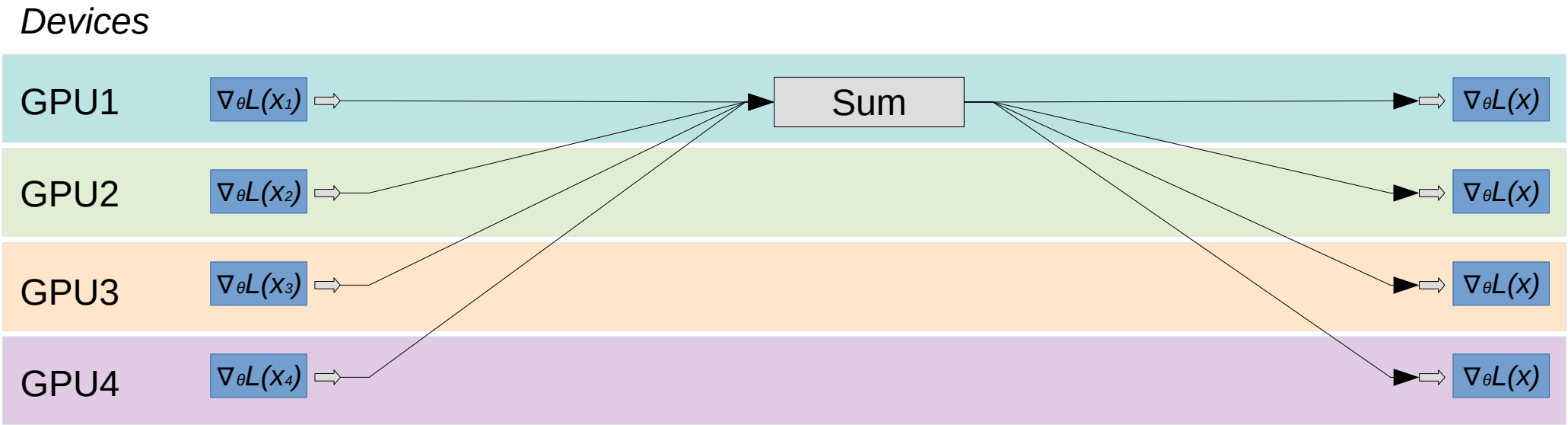
Naive implementation



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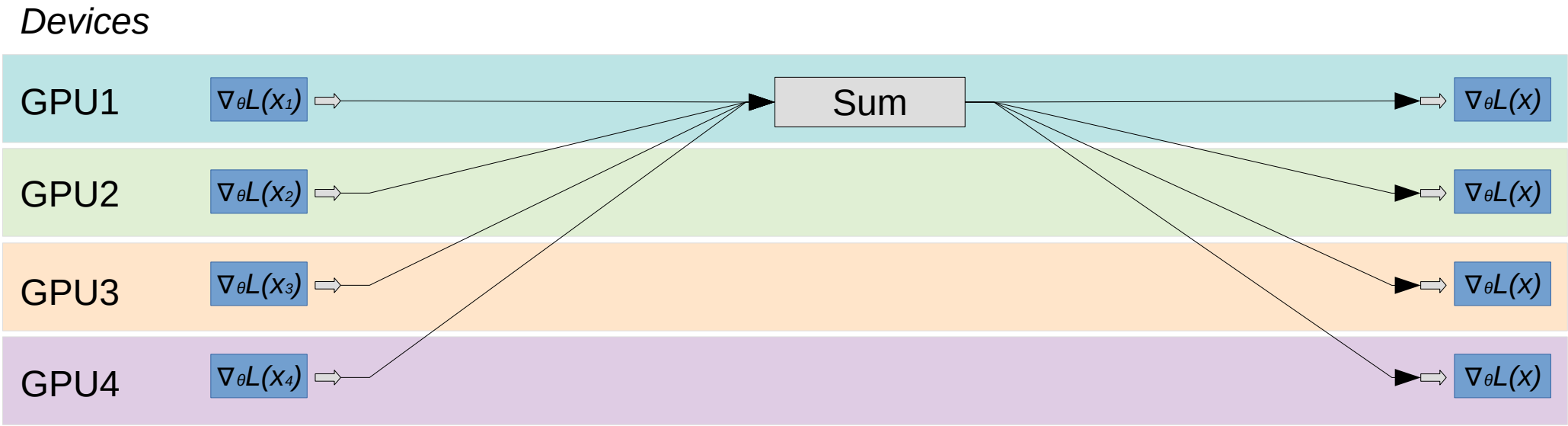
Q: Can we do better?



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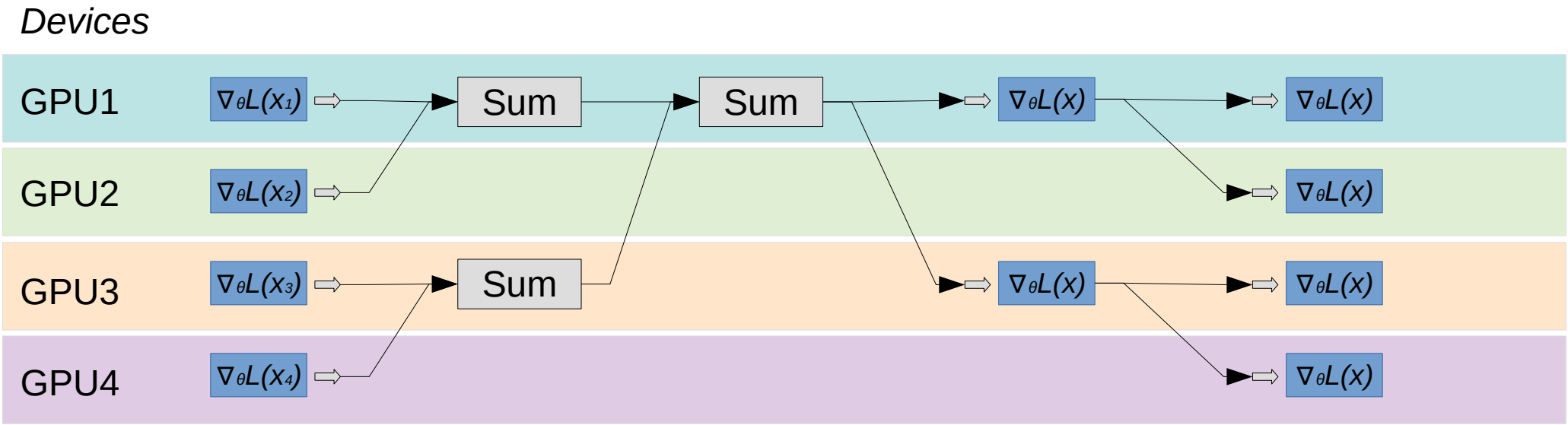
Tree-allreduce



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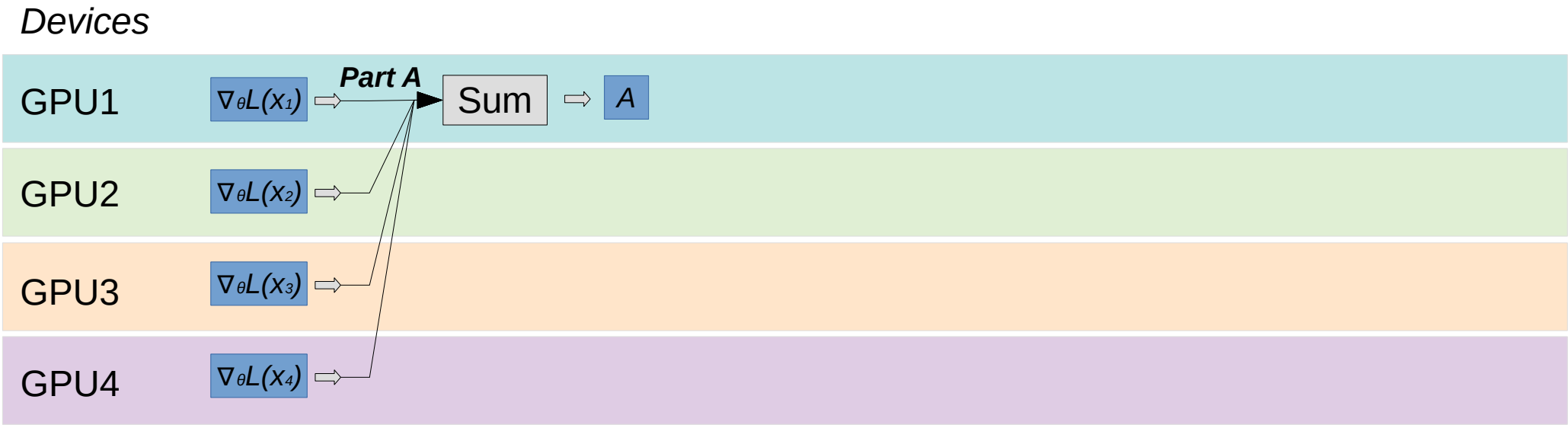
Butterfly-allreduce – split data into chunks (ABCD)



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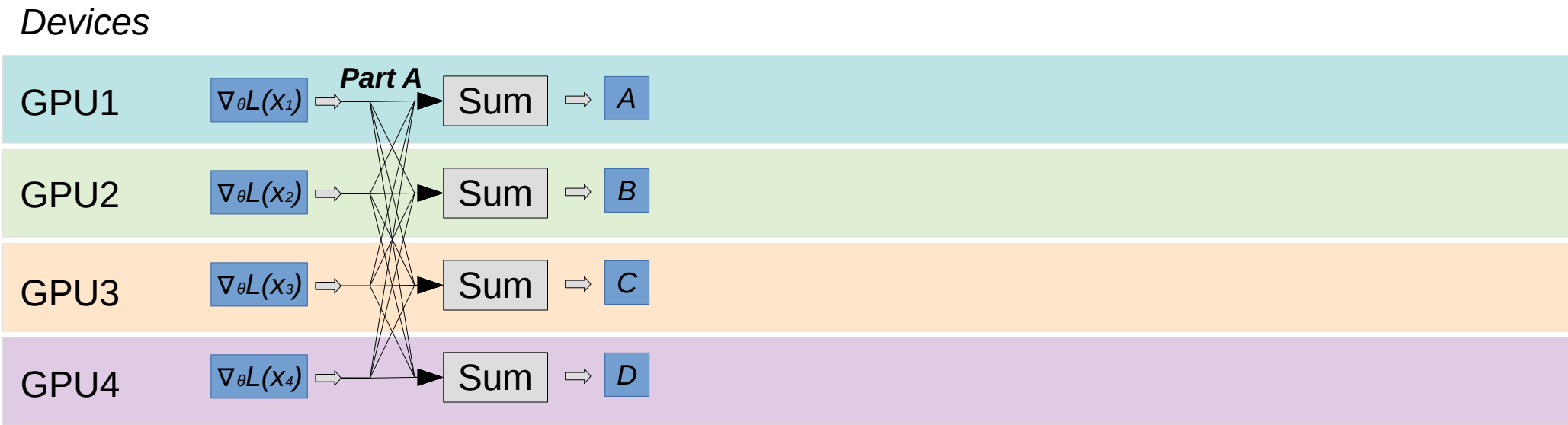

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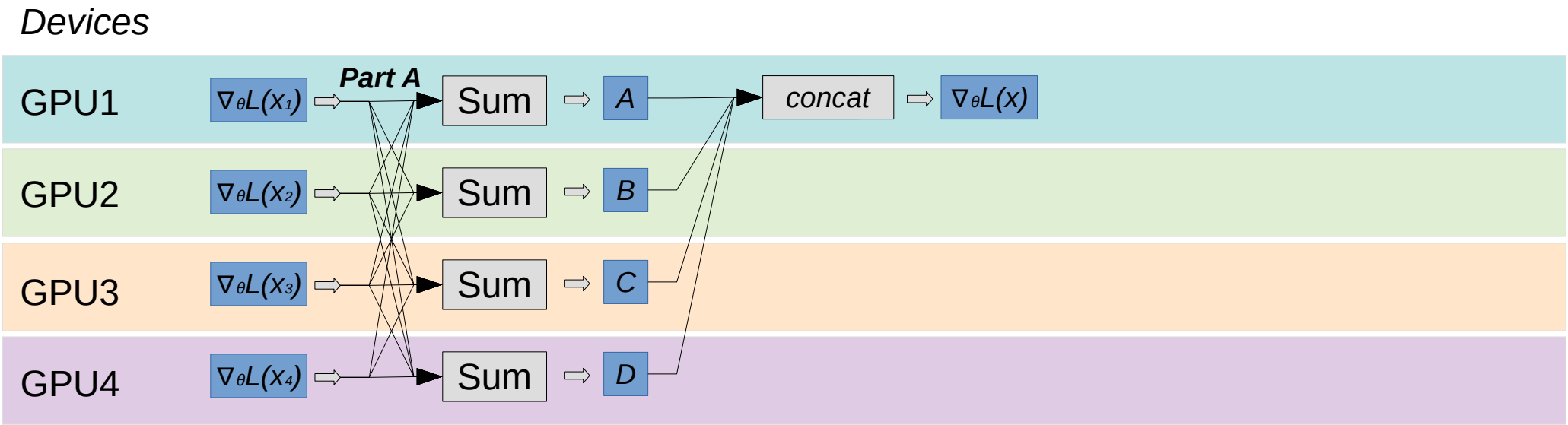

Faster allreduce

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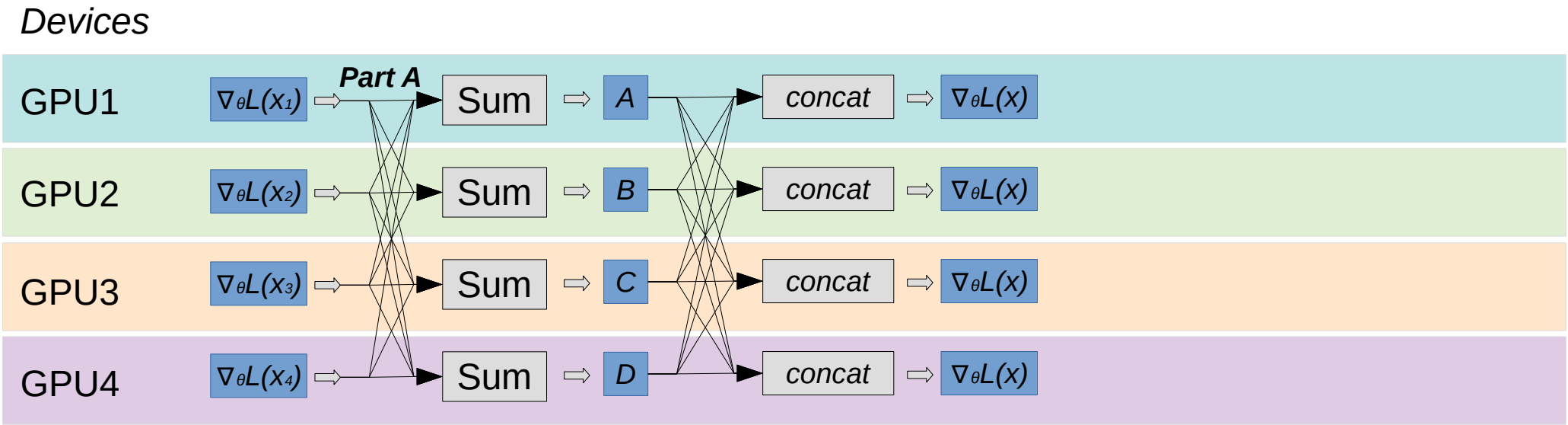
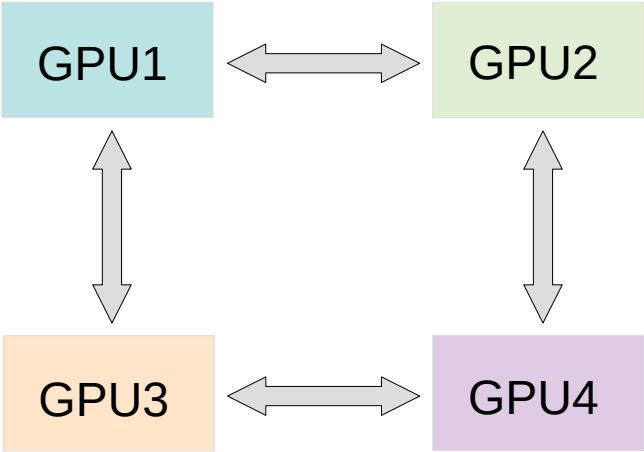
Ring-allreduce – split data into chunks (ABCD)



Ring allreduce

Bonus quest: you can only send data between **adjacent** gpus



Ring topology

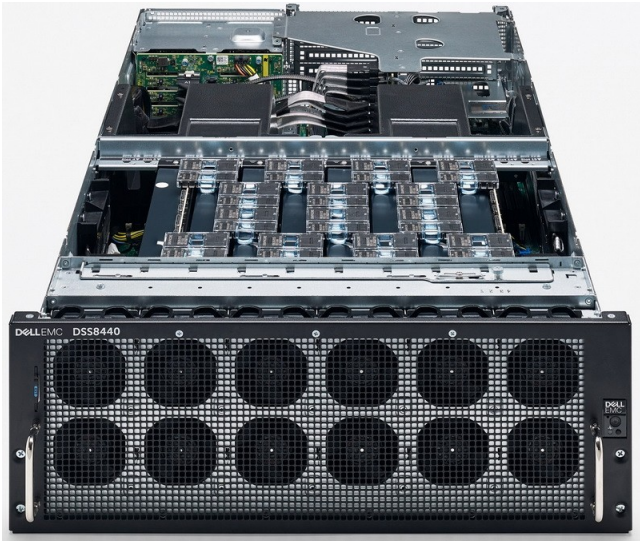


Image: graphcore IPU server

Answer & more: tinyurl.com/ring-allreduce-blog

Ring allreduce

Bonus quest: you can only send data between **adjacent** gpus

[Time to use the whiteboard]

Answer & more: tinyurl.com/ring-allreduce-blog

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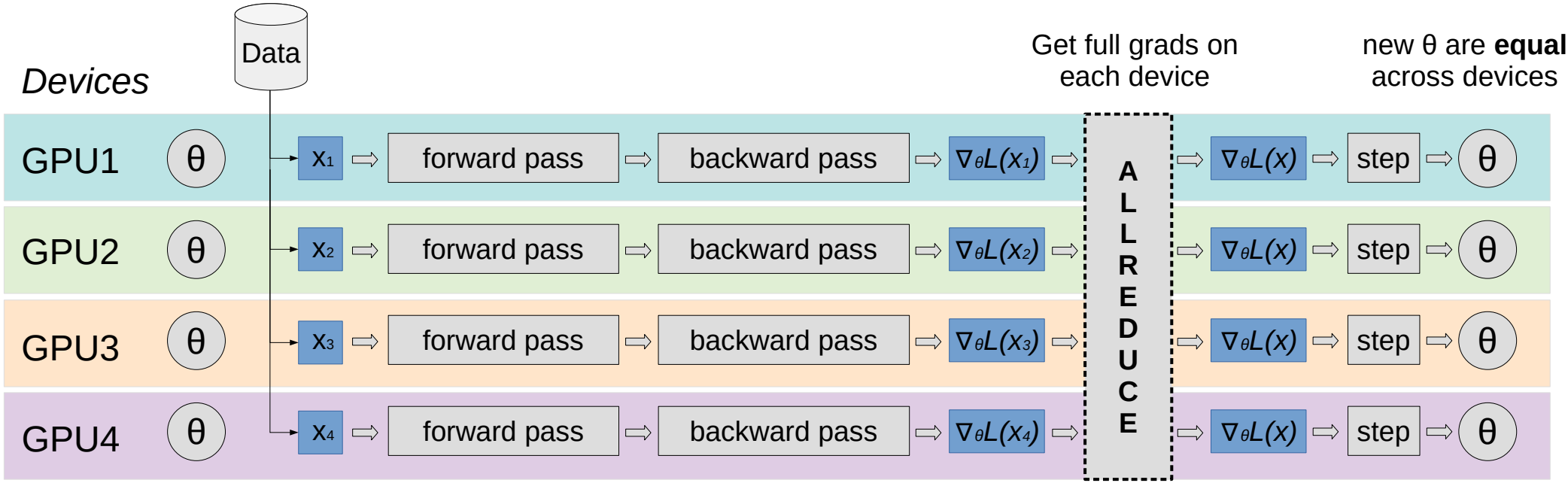


</Data-parallel>

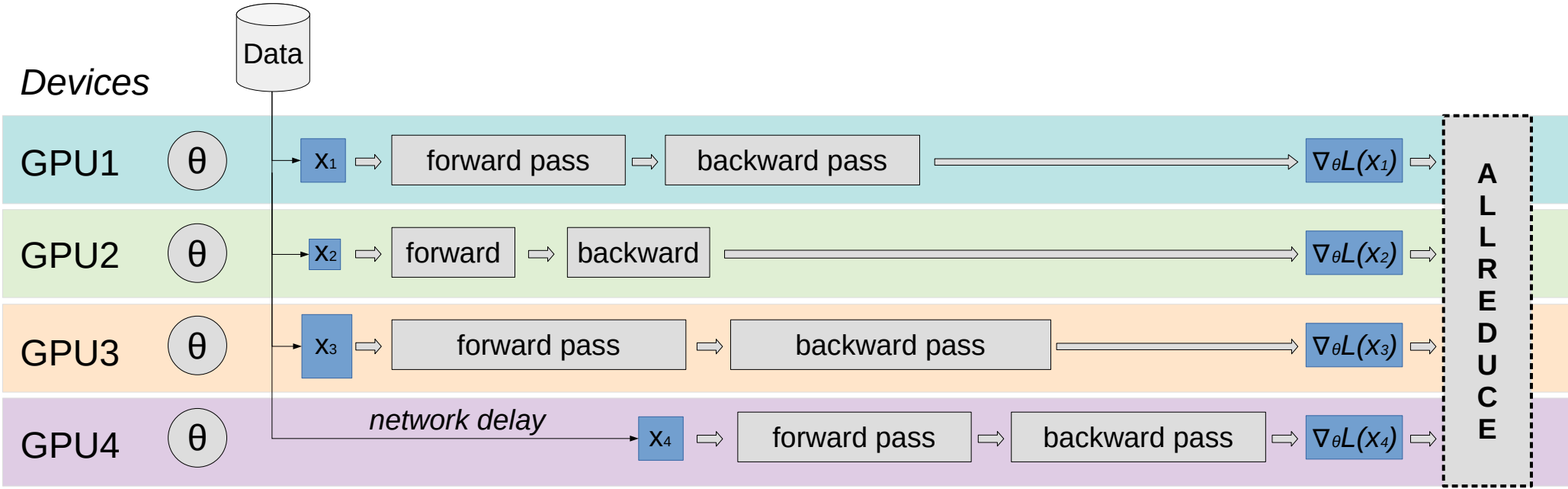
+ easy to implement
+ can scale to 100s of gpus
+ can be fault-tolerant
- model must fit in 1 gpu
- large batches aren't always good for generalization

- 2-4 GPUs & no time – naive data parallel tinyurl.com/torch-data-parallel
- 4+ GPUs or multiple hosts – distributed (allreduce) github.com/horovod/horovod
  - Intro to pytorch distributed: tinyurl.com/distributed-dp **or in 15 minutes!**
- Somewhat faulty GPU/network: synchronous data parallel + drop stragglers
- Very faulty or uneven resources: asynchronous data parallel (more later)
- Efficient training with large batches: LAMB https://arxiv.org/abs/1904.00962
- Dynamically adding or removing resources: https://tinyurl.com/torch-elastic

Decentralized training vs real-world tasks

arxiv.org/abs/1706.02677

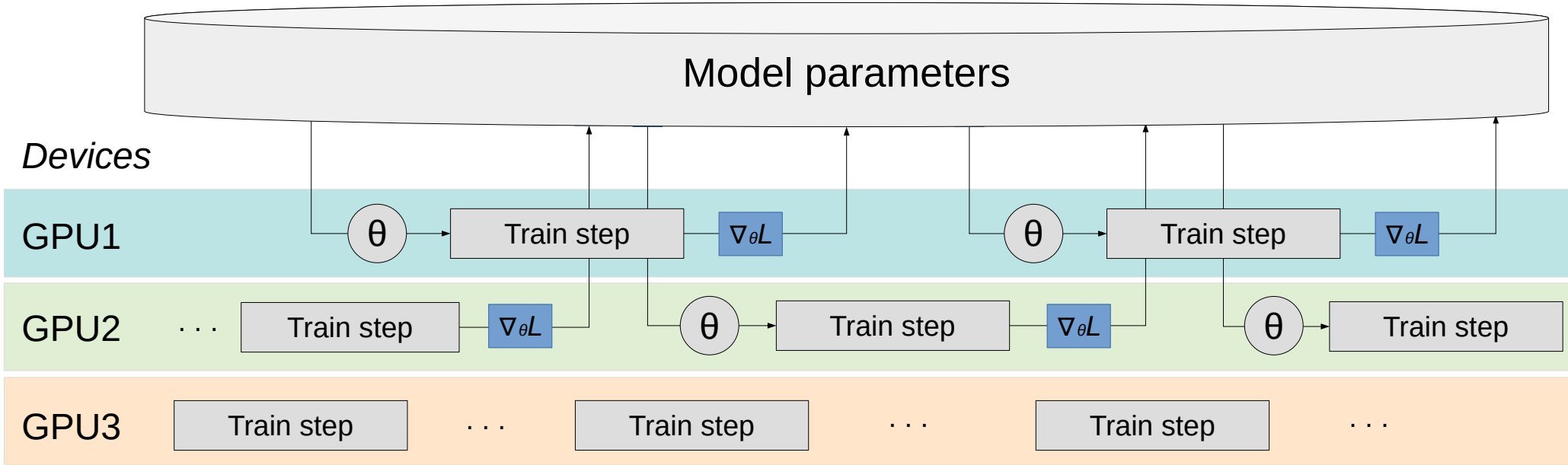
Each gpu has different processing time & delays
Q: can we improve device utilization?



Recap: Parameter Server

HOGWILD! arxiv.org/abs/1106.5730

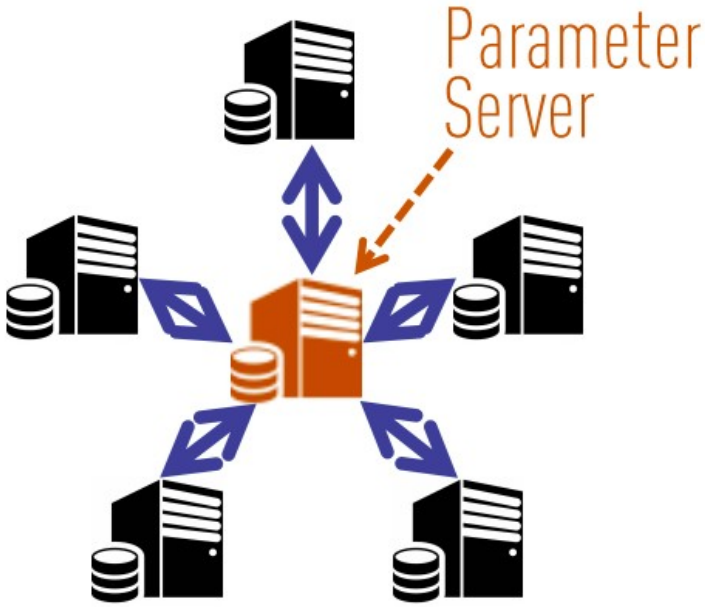
Idea: remove synchronization step alltogether, use parameter server



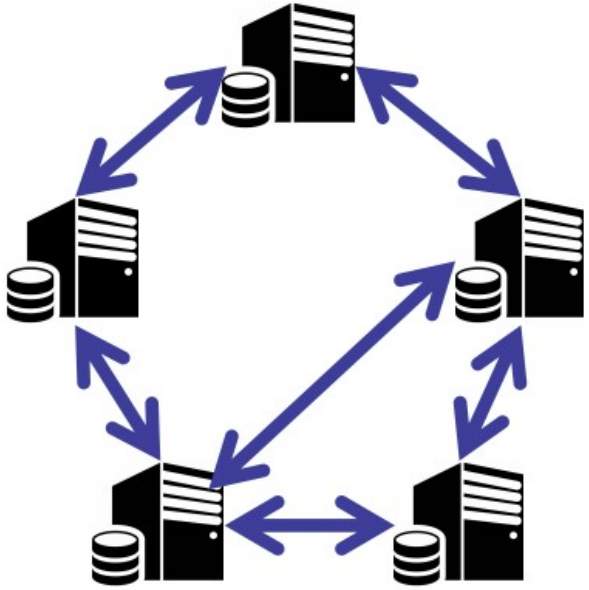
Problem: parameter servers need to ingest tons of data over training

Decentralized Training with Gossip

Gossip (communication): https://tinyurl.com/boyd-gossip-2006
Gossip outperforms All-Reduce: https://tinyurl.com/can-dsgd-outperform



(a) Centralized Topology



(b) Decentralized Topology

Decentralized Training with Gossip

Source: https://tinyurl.com/can-dsgd-outperform

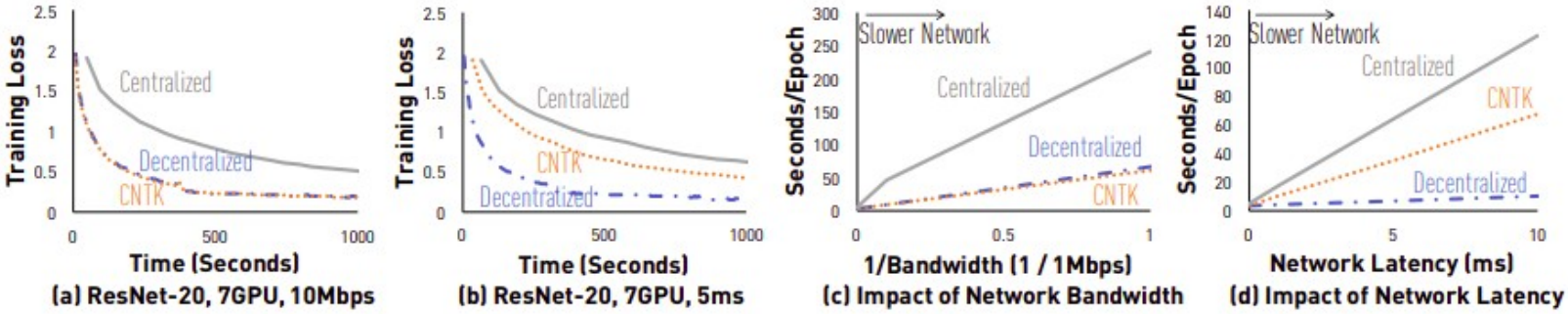


Figure 2: Comparison between D-PSGD and two centralized implementations (7 and 10 GPUs).

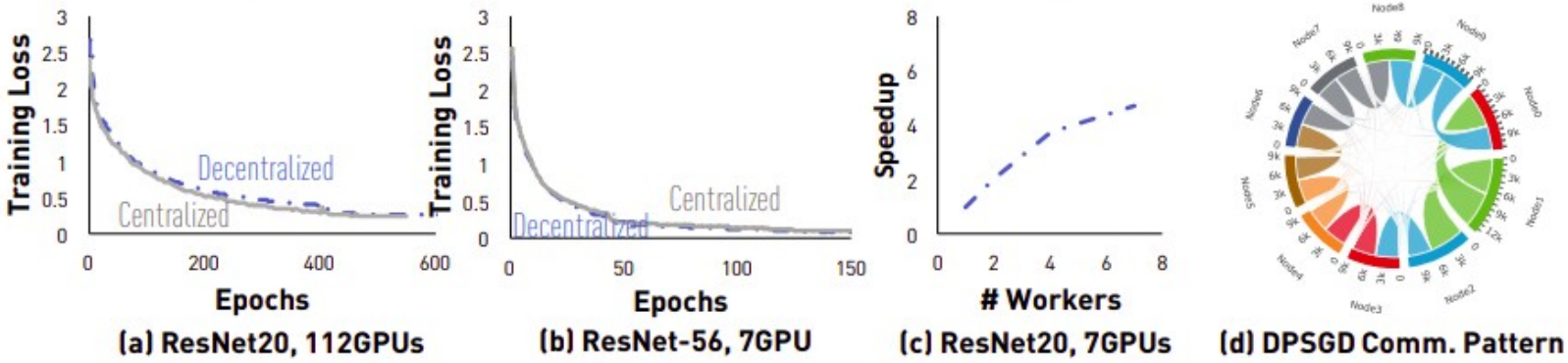
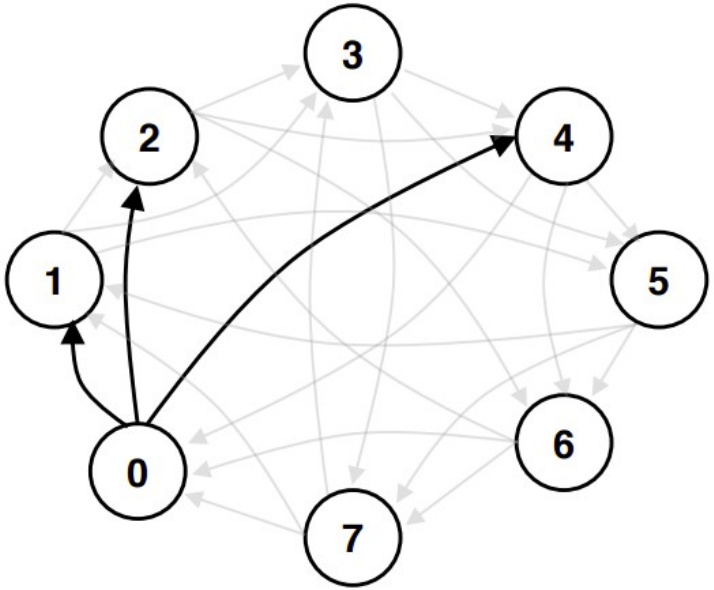


Figure 3: (a) Convergence Rate; (b) D-PSGD Speedup; (c) D-PSGD Communication Patterns.

Stochastic Gradient Push

Source: https://arxiv.org/abs/1811.10792



(a) Directed Exponential Graph highlighting node 0's out-neighbours

Stochastic Gradient Push

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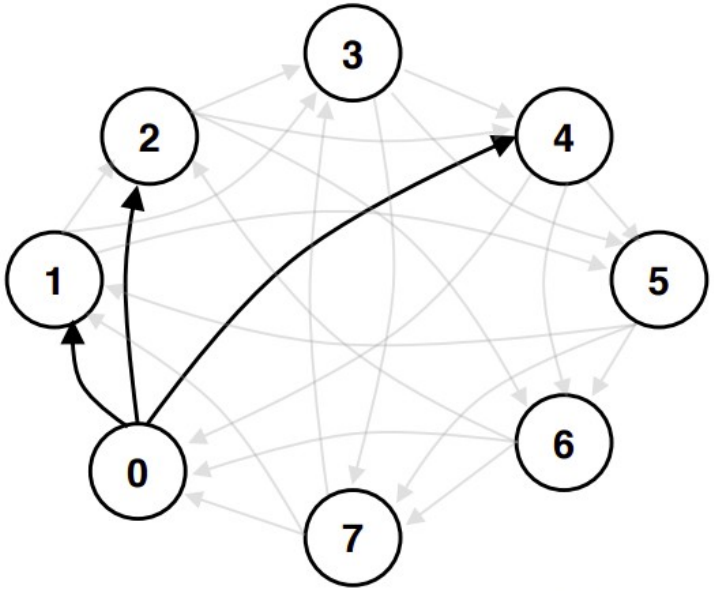


(a) Directed Exponential Graph highlighting node 0's out-neighbours

Algorithm 1 Stochastic Gradient Push (SGP)

Require: Initialize $\gamma > 0$, $\boldsymbol{x}_i^{(0)} = \boldsymbol{z}_i^{(0)} \in \mathbb{R}^d$ and $w_i^{(0)} = 1$ for all nodes $i \in \{1, 2, \ldots, n\}$

1: **for** $k = 0, 1, 2, \cdots, K$, at node i, **do**

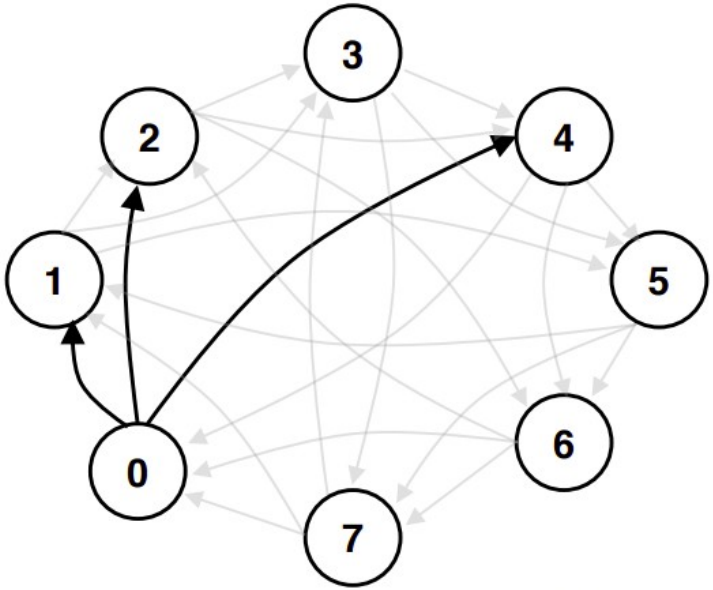
2: Sample new mini-batch $\xi_i^{(k)} \sim \mathcal{D}_i$ from local distribution

3: Compute mini-batch gradient at $\boldsymbol{z}_i^{(k)}$: $\nabla \boldsymbol{F}_i(\boldsymbol{z}_i^{(k)}; \xi_i^{(k)})$

<to be continued>

Stochastic Gradient Push

Source: https://arxiv.org/abs/1811.10792



(a) Directed Exponential Graph highlighting node 0's out-neighbours

Algorithm 1 Stochastic Gradient Push (SGP)

Require: Initialize $\gamma > 0$, $\boldsymbol{x}_i^{(0)} = \boldsymbol{z}_i^{(0)} \in \mathbb{R}^d$ and $w_i^{(0)} = 1$ for all nodes $i \in \{1, 2, \ldots, n\}$

1: **for** $k = 0, 1, 2, \cdots, K$, at node i, **do**

2: Sample new mini-batch $\xi_i^{(k)} \sim \mathcal{D}_i$ from local distribution

3: Compute mini-batch gradient at $\boldsymbol{z}_i^{(k)}$: $\nabla \boldsymbol{F}_i(\boldsymbol{z}_i^{(k)}; \xi_i^{(k)})$

4: $\boldsymbol{x}_i^{(k+\frac{1}{2})} = \boldsymbol{x}_i^{(k)} - \gamma \nabla \boldsymbol{F}_i(\boldsymbol{z}_i^{(k)}; \xi_i^{(k)})$

normal GD step

<to be continued>

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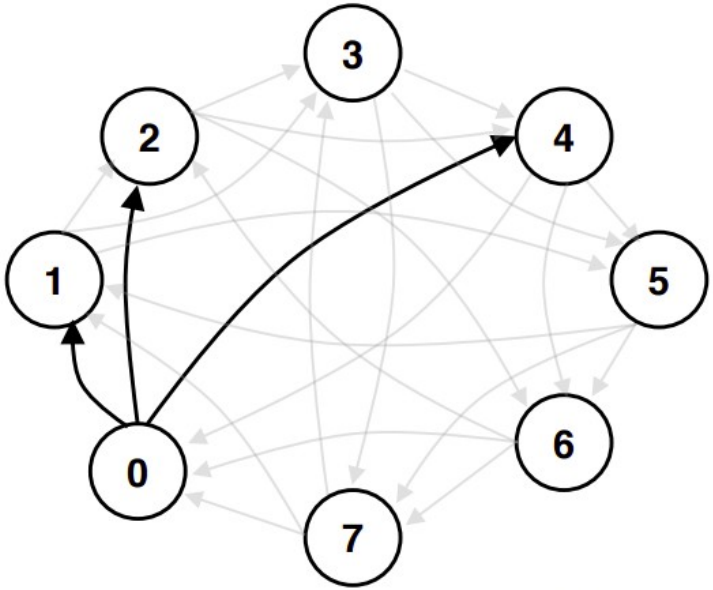

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5: Send $\left(p_{j,i}^{(k)} \boldsymbol{x}_i^{(k+\frac{1}{2})}, p_{j,i}^{(k)} w_i^{(k)}\right)$ to out-neighbors;

receive $\left(p_{i,j}^{(k)} \boldsymbol{x}_j^{(k+\frac{1}{2})}, p_{i,j}^{(k)} w_j^{(k)}\right)$ from in-neighbors

<to be continued>

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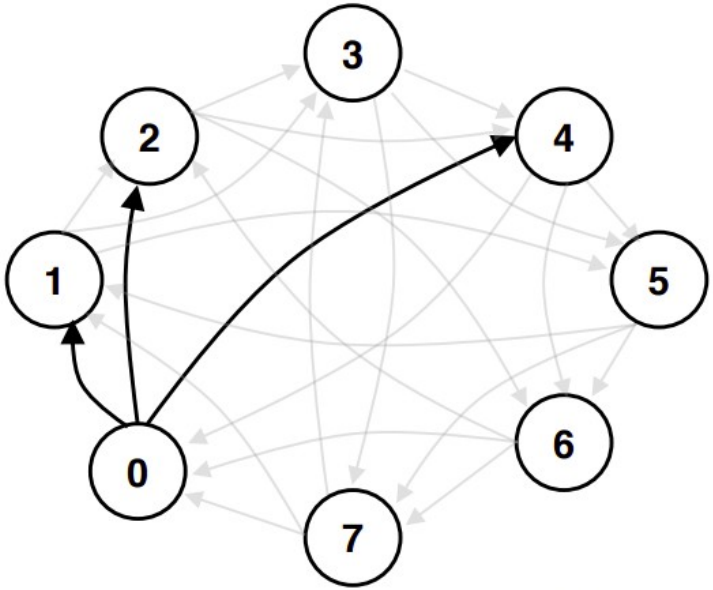

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  $\boldsymbol{x}_i^{(k+1)} = \sum_j p_{i,j}^{(k)} \boldsymbol{x}_j^{(k+\frac{1}{2})}$

  $w_i^{(k+1)} = \sum_j p_{i,j}^{(k)} w_j^{(k)}$

  $\boldsymbol{z}_i^{(k+1)} = \boldsymbol{x}_i^{(k+1)} / w_i^{(k+1)}$

end for

weighted average

Stochastic Gradient Push

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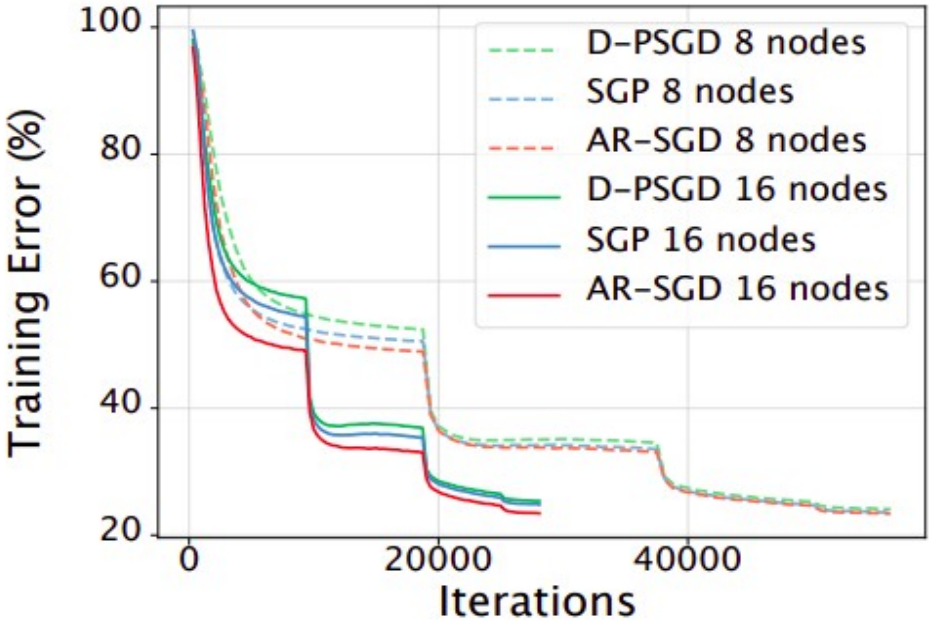
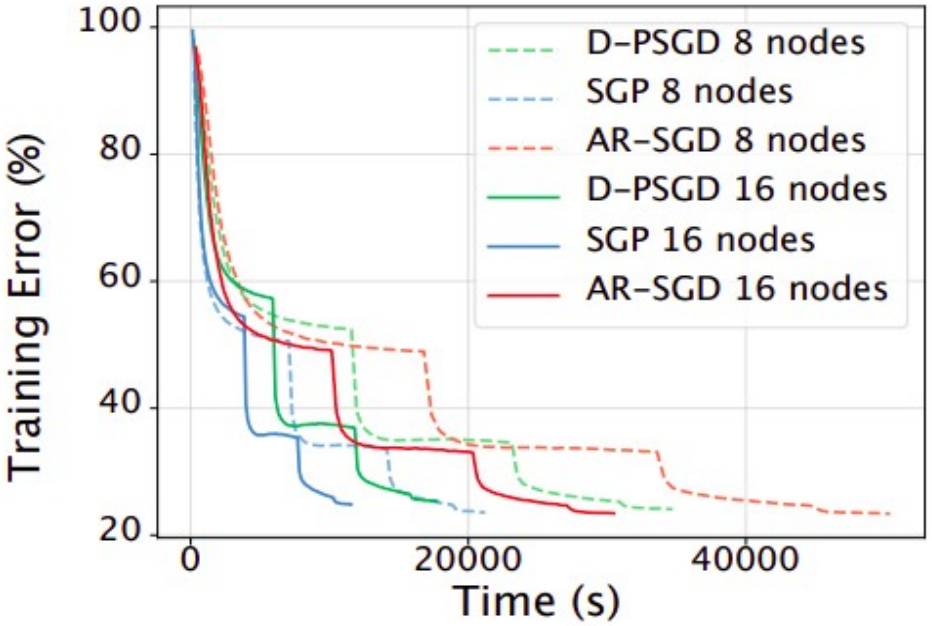
SGP vs ImageNet (ResNet50 + SGD w/ momentum)





Stochastic Gradient Push

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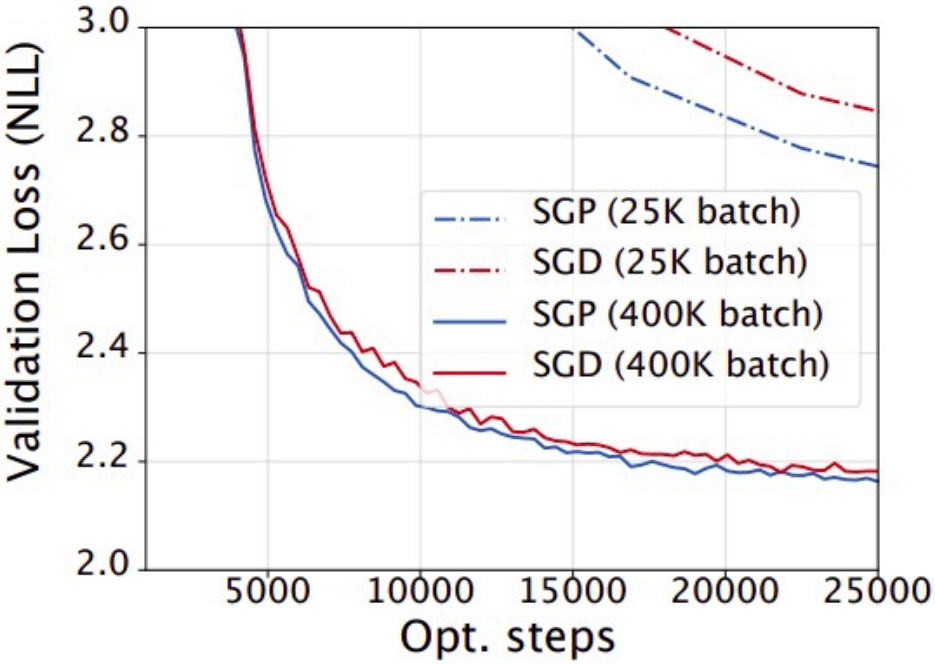
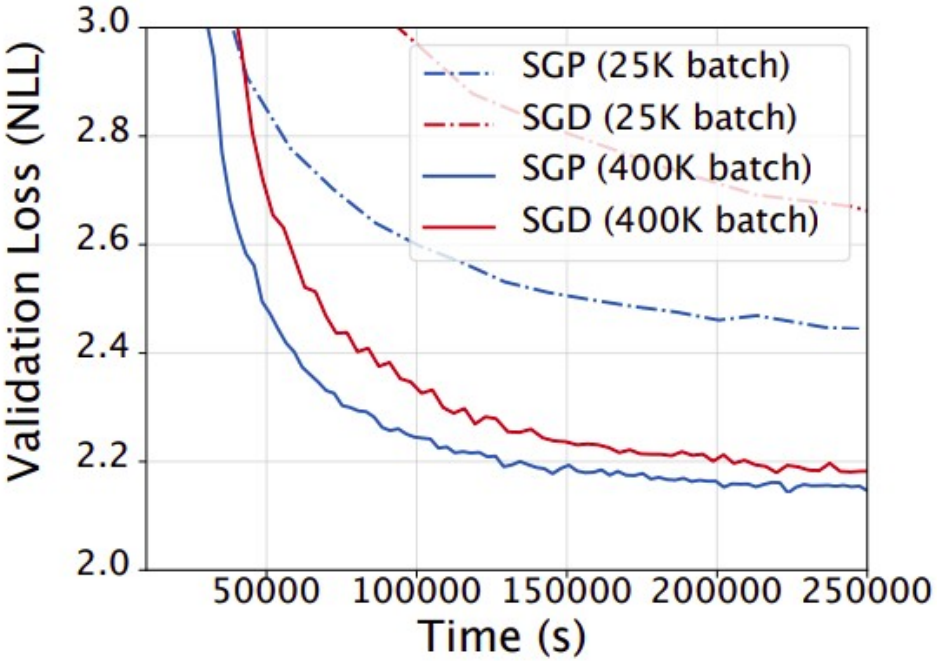
SGP vs WMT English-German (Transformer, Adam)





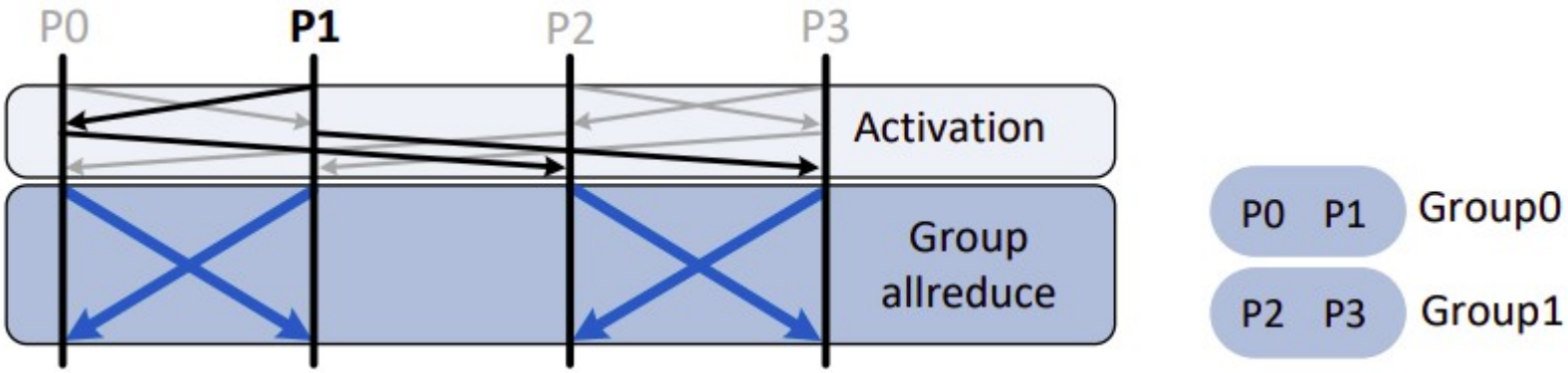
Gossip vs All-Reduce

Your thoughts?

Gossip + All-Reduce

Source: arxiv.org/abs/2005.00124

Core idea: run all-reduce in independent groups
You only have to synchronize for your small group
Swap groupmates between iterations



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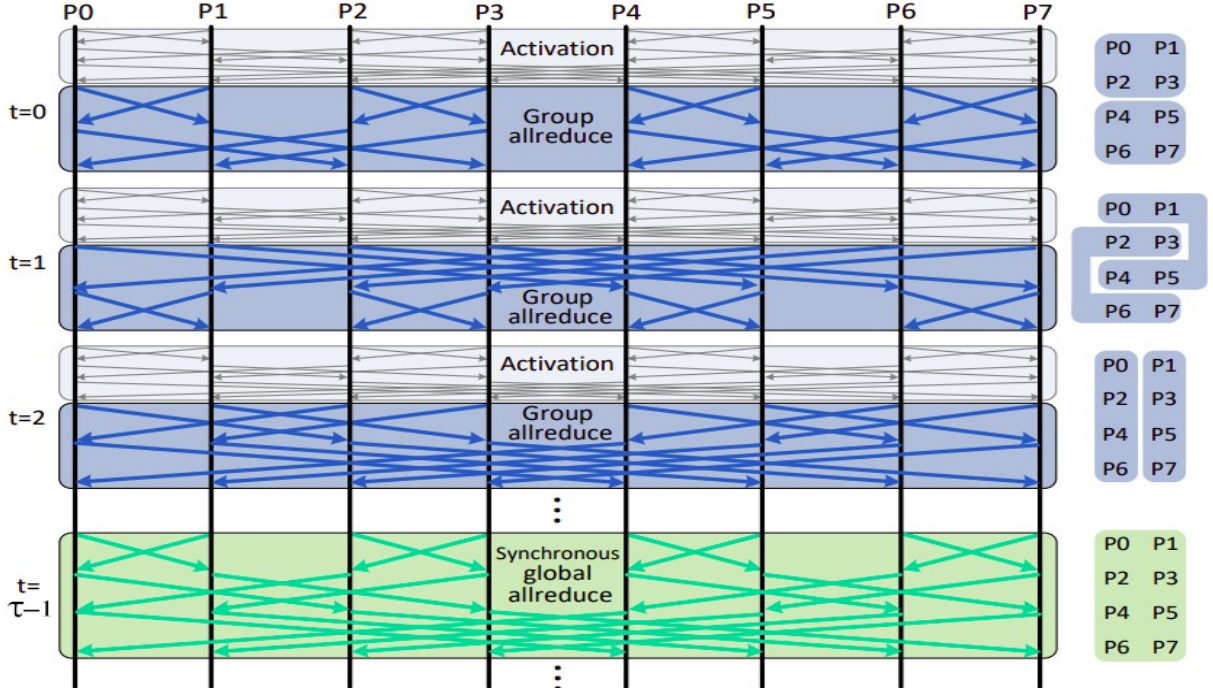
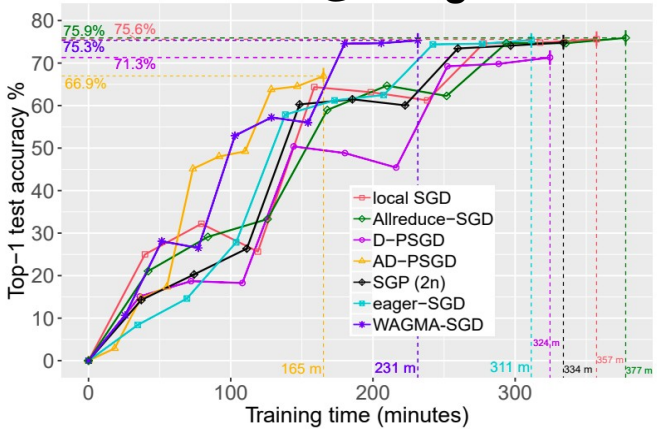


Gossip + All-Reduce

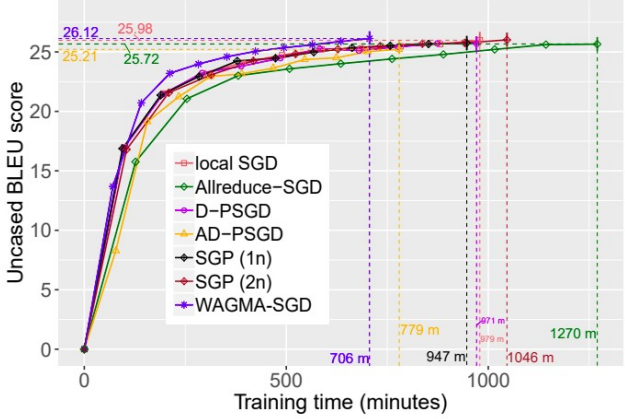
Source: arxiv.org/abs/2005.00124

Experiment setup: up to 1024 GPU,
Natural (or emulated) network latency

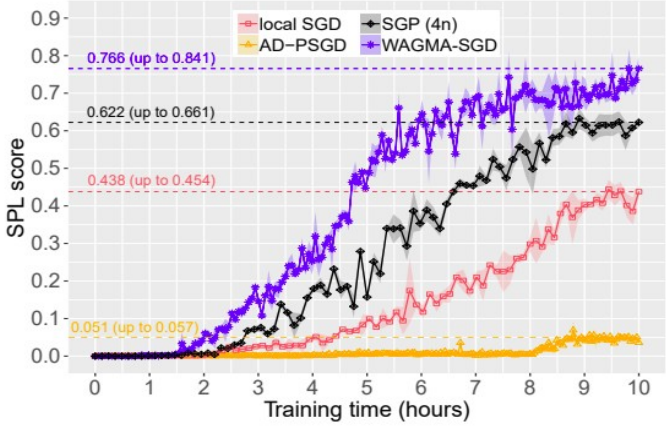
Image Classification
ResNet50 @ ImageNet



Machine Translation
Transformer @ WMT17



Reinforcement Learning
DDPO on Habitat



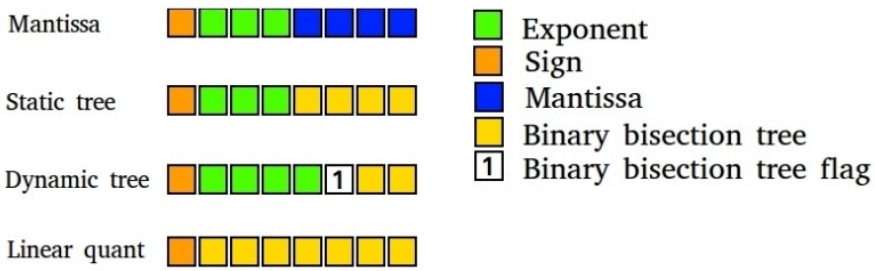
Q: what if sending tensors during AllReduce takes too long?

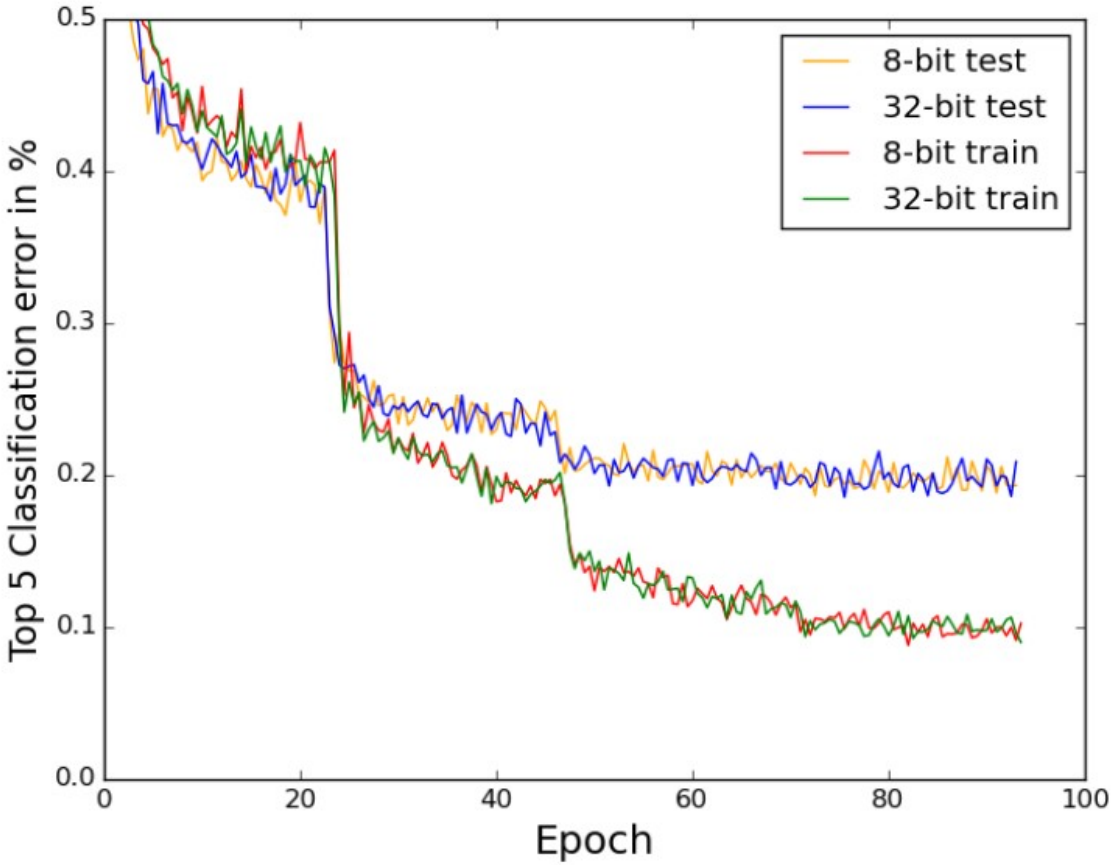
Quantized communication

https://arxiv.org/abs/1511.04561

TL;DR
- send data in 8-bit
- all computations in 32-bit
- choose best data format

PROFIT: same quality as float16





Can we compress further?
without losing quality

Error Feedback + PowerSGD

https://arxiv.org/abs/1901.09847 - EF theory
https://arxiv.org/abs/1905.13727 - PowerSGD

TL;DR - use extreme compression, e.g. 1-bit or top-5% gradients

- if you lose something in compression, **reuse it on the next step**

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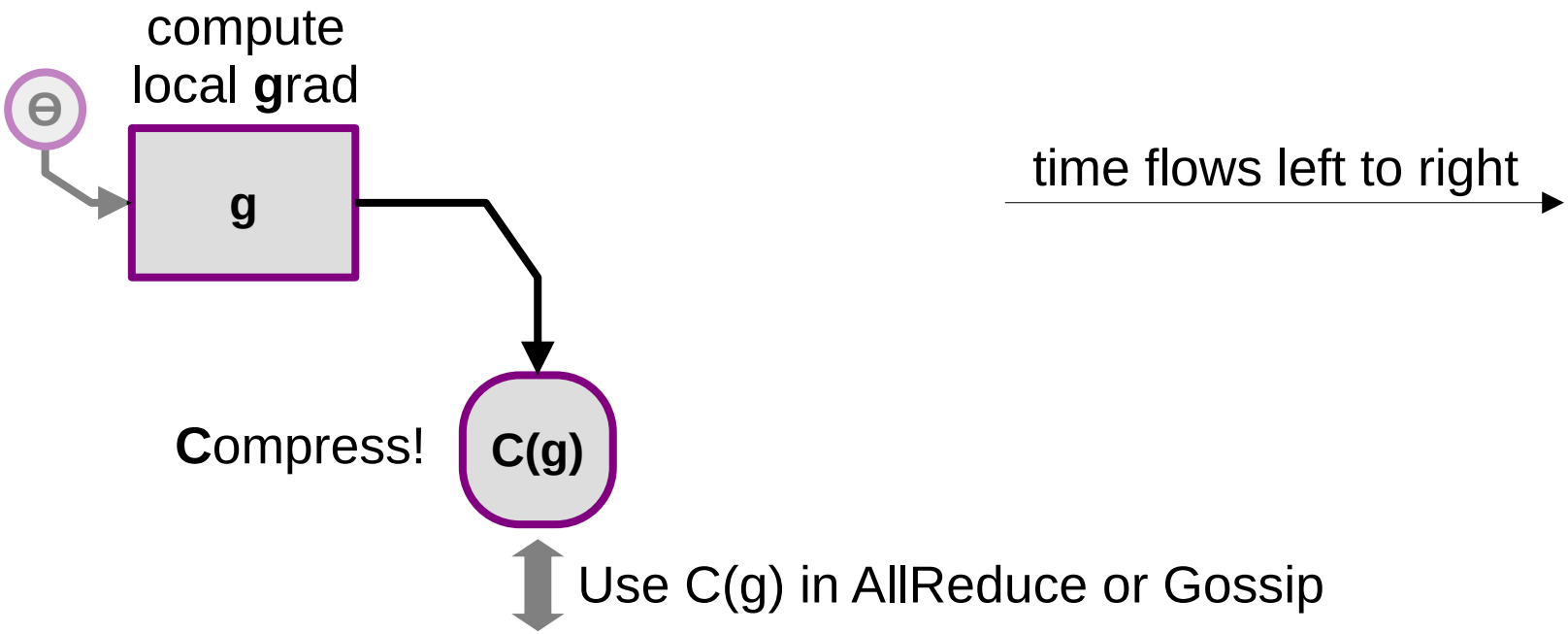

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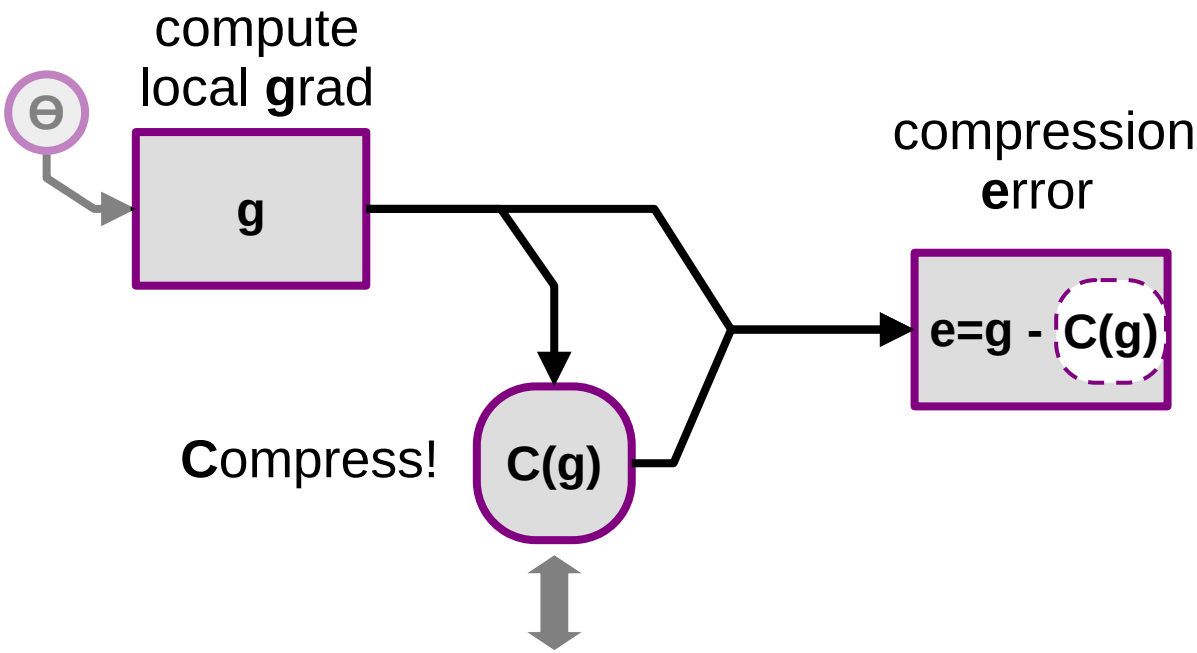

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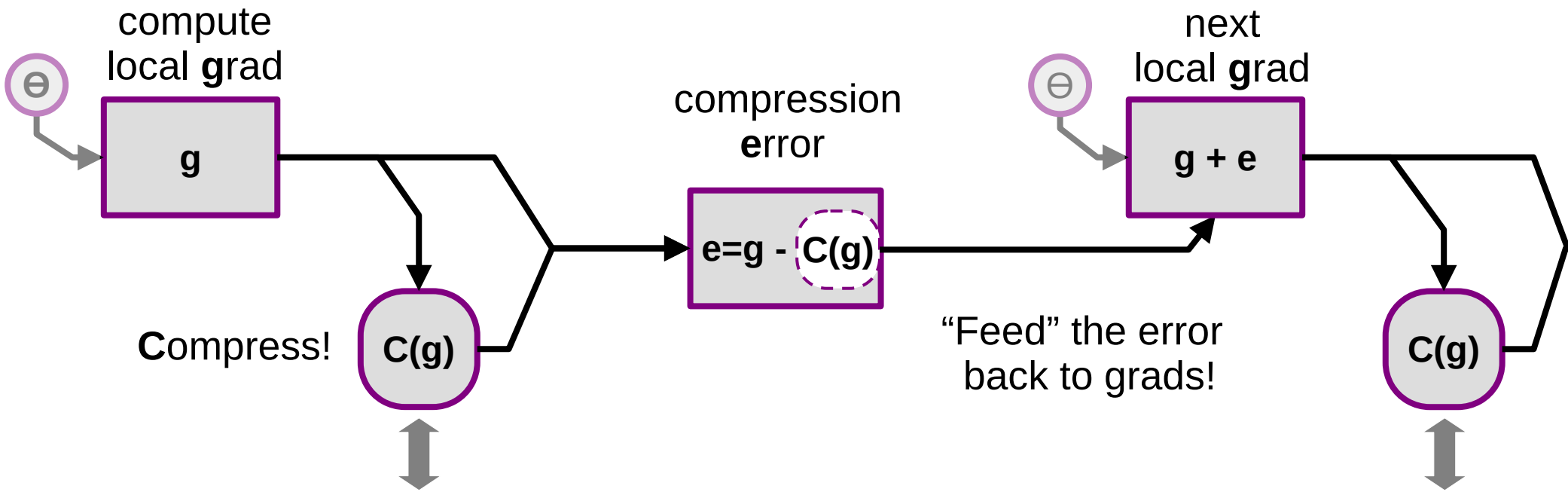

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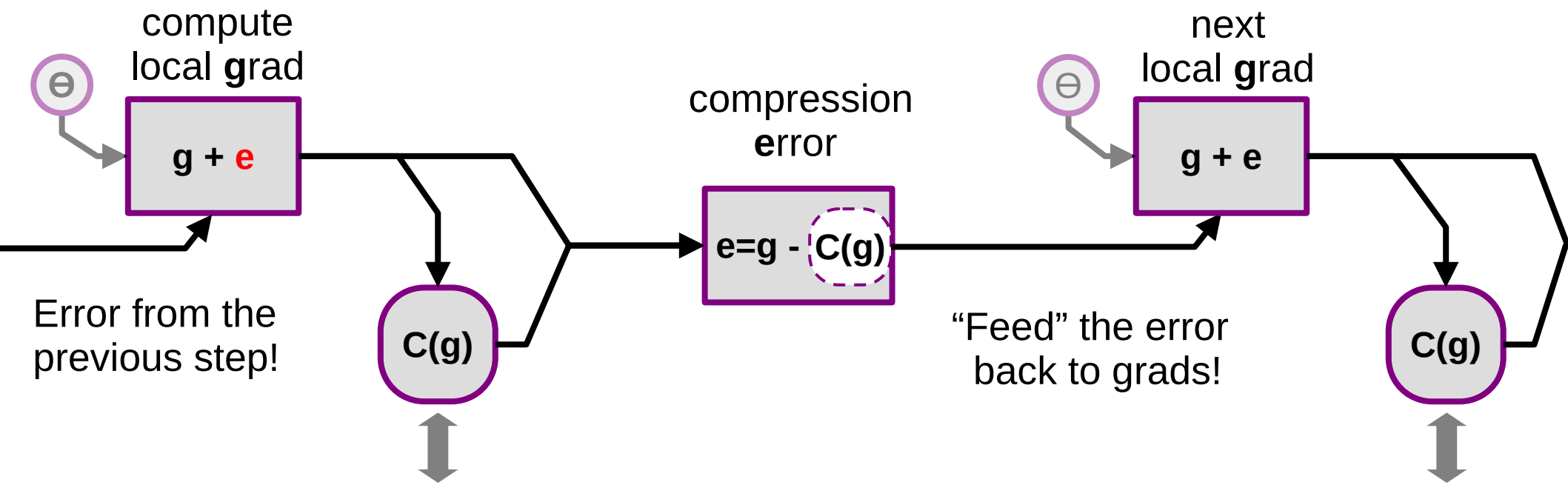

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hyperparameters: learning rate γ, momentum parameter λ
initialize model parameters $\mathbf{x} \in \mathbb{R}^d$, momentum $\mathbf{m} \leftarrow \mathbf{0} \in \mathbb{R}^d$, replicated across workers
at each worker $w = 1, \ldots, W$ **do**
  initialize memory $\mathbf{e}_w \leftarrow \mathbf{0} \in \mathbb{R}^d$
  for each iterate $t = 0, \ldots$ **do**
    Compute a stochastic gradient $\mathbf{g}_w \in \mathbb{R}^d$.
    $\Delta_w \leftarrow \mathbf{g}_w + \mathbf{e}_w$ ▷ Incorporate error-feedback into update
    $\mathcal{C}(\Delta_w) \leftarrow \textsc{compress}(\Delta_w)$
    $\mathbf{e}_w \leftarrow \Delta_w - \textsc{decompress}(\mathcal{C}(\Delta_w))$ ▷ Memorize local errors
    $\mathcal{C}(\Delta) \leftarrow \textsc{aggregate}(\mathcal{C}(\Delta_1), \ldots, \mathcal{C}(\Delta_W))$ ▷ Exchange gradients
    $\Delta' \leftarrow \textsc{decompress}(\mathcal{C}(\Delta))$ ▷ Reconstruct an update $\in \mathbb{R}^d$
    $\mathbf{m} \leftarrow \lambda \mathbf{m} + \Delta'$
    $\mathbf{x} \leftarrow \mathbf{x} - \gamma\,(\Delta' + \mathbf{m})$
  end for
end at

PowerSGD: low-rank approx grads + Error Feedback

https://arxiv.org/abs/1901.09847 - EF theory
https://arxiv.org/abs/1905.13727 - PowerSGD

The update vector Δ_w is treated as a list of tensors corresponding to individual model parameters. Vector-shaped parameters (biases) are aggregated uncompressed. Other parameters are reshaped into matrices. The functions below operate on such matrices independently. For each matrix $M \in \mathbb{R}^{n \times m}$, a corresponding $Q \in \mathbb{R}^{m \times r}$ is initialized from an i.i.d. standard normal distribution.

function COMPRESS+AGGREGATE(update matrix $M \in \mathbb{R}^{n \times m}$, previous $Q \in \mathbb{R}^{m \times r}$)
  $P \leftarrow MQ$
  $P \leftarrow$ ALL REDUCE MEAN(P) ▷ Now, $P = \frac{1}{W}(M_1 + \ldots + M_W)Q$
  $\hat{P} \leftarrow$ ORTHOGONALIZE(P) ▷ Orthonormal columns
  $Q \leftarrow M^\top \hat{P}$
  $Q \leftarrow$ ALL REDUCE MEAN(Q) ▷ Now, $Q = \frac{1}{W}(M_1 + \ldots + M_W)^\top \hat{P}$
  return the compressed representation $(\hat{P}, Q)$.
end function
function DECOMPRESS($\hat{P} \in \mathbb{R}^{n \times r}$, $Q \in \mathbb{R}^{m \times r}$)
  return $\hat{P}Q^\top$
end function

Read More: gradient compression

https://arxiv.org/abs/1901.09847 - EF theory

https://arxiv.org/abs/1905.13727 - PowerSGD

https://arxiv.org/abs/2106.05203 - better EF'21

https://arxiv.org/abs/2110.03294 - more EF'21

```
import torch.distributed.algorithms.ddp_comm_hooks.powerSGD_hook as powerSGD

ddp_model = nn.parallel.DistributedDataParallel(
    module=model,
    device_ids=[rank],
)

state = PowerSGD.PowerSGDState(
  process_group=process_group,
  matrix_approximation_rank=1,
  start_powerSGD_iter=1_000,
)
ddp_model.register_comm_hook(state, PowerSGD.powerSGD_hook)
```

„That's all Folks!"
Isberg©

Summary: operation parallelism

Data-parallel: ***???***

Model-parallel: ***???***

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Data-parallel: *one process applies all model on* ***partial data***

best for smaller model, more computations

Model-parallel: *one process applies* ***partial model*** *on all data*

best for larger model, fewer computations

Which one is better..

for ResNet50?

for Llama 70B?

In general?

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~~*Which one is better..*~~

~~*for ResNet50?*~~

~~*In Llama 70B?*~~

It depends…

- on model size

- on compute